



M U L 2 \_ N E W

# User Guide

Building models, reference systems, running  
analyses and reading results

Politecnico di Torino · MUL2 project  
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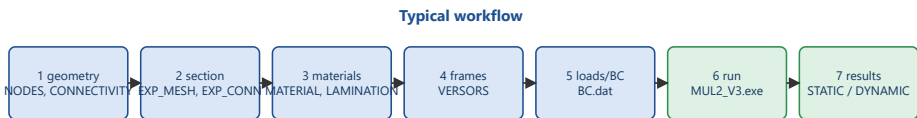
# Getting started

## 1.1 What MUL2\_NEW does

MUL2\_NEW analyses linear elastic **beams, plates and solids**, alone or mixed in the same model, with a family of refined structural theories (the Carrera Unified Formulation). It answers two questions:

1. **Static analysis (101)**: how does the structure deform and what are the strains and stresses under point loads and supports?
2. **Free vibration (103)**: what are its natural frequencies and mode shapes?

You describe the structure with a handful of text files, run one executable and read text, VTK and GMSH files. The workflow is:



**Figure 1.1.** Typical workflow.

## 1.2 Requirements

- Windows 10/11 (64-bit).
- The Intel oneAPI runtime libraries (MKL and the Fortran/OpenMP runtime) must be found at run time: either install oneAPI or put its `mk1\...\bin` and `compiler\...\bin` directories in the `PATH` (the web interface does this for you).
- For visualisation: ParaView (VTK files) or GMSH ( `.msh` files). Any text editor to edit the inputs.

The executable is `MUL2_V3.exe`. If you do not have it, build it as described in the Implementation Guide ( `CREATE_VS_SOLUTION.cmd build` ).

## 1.3 Your first run in two minutes

The folder `DOC/BOOKS/examples/cases` contains ready-to-run inputs. Take the cantilever beam with Taylor order 2:

```
mkdir run1 & cd run1
mkdir STATIC DYNAMIC WORK REPORT
xcopy /E /I ..\DOC\BOOKS\examples\cases\beam_te2_static\INPUT INPUT
..\BUILD\WINDOWS_IFX\Release\MUL2_V3.exe INPUT
```

The console prints something like

```
[INFO] MUL2_V3 RUNTIME INITIALIZED
[INFO] INPUT DIRECTORY: INPUT
[INFO] MODEL: 16 NODES, 5 ELEMENTS, SOLUTION 101
[INFO] PREPROCESSING: 288 DOF, 180 GAUSS POINTS
[INFO] K ASSEMBLED: 288 DOF, 24624 NNZ,      0.012 S
[INFO] STATIC SOLVE:      0.096 S, MAX |U| =  8.4935E-04
[INFO] WRITTEN STATIC/RESULTS PARA_01.vtk
[INFO] TIMES [S] INPUT/PRE/ANALYSIS/OUT:    0.005    0.000    0.108    0.032
[INFO] CPU [S]:      0.171875 WALL [S]:      0.223000
```

### NOTE

$\max |U|$  is the largest *coefficient* of the solution vector. With a Taylor expansion the coefficients of the higher terms are not displacements of a point, so this number is only a sanity check (not the tip deflection). Read displacements from `POST_POINT.dat`.

The results are in `STATIC/POST_POINT.dat` (values at the requested points), `STATIC/RESULTS_PARA_01.vtk` (the field, to open in ParaView) and `REPORT/WARNING_file.dat` (warnings, if any). The exit code is 0 when everything went well.

### NOTE

The program writes its outputs in the **current directory**, in the folders `STATIC`, `DYNAMIC`, `WORK` and `REPORT` (it creates them if missing). The input directory can be given as the first argument; without an argument the program reads the name of the input directory from the first line of the file `PATH_input.dat` and, if that file does not exist, uses `INPUT`.

# 1.4 Command line

```
MUL2_V3.exe [input_directory]
MUL2_V3.exe --version
```

Environment variables that change the behaviour:

**Table 1.1.** Environment variables.

| Variable            | Effect                                                                               |
|---------------------|--------------------------------------------------------------------------------------|
| OMP_NUM_THREADS     | number of threads used for the assembly and the output (results do not depend on it) |
| MUL2_DEBUG          | 1 prints extra debugging lines                                                       |
| MUL2_GENERAL_KERNEL | 1 uses the slower reference element kernel (for checks)                              |

# 1.5 What is in a run directory

|                |                                                      |
|----------------|------------------------------------------------------|
| my_run/        |                                                      |
| INPUT/         | your input files (see Chapter 4)                     |
| PATH_input.dat | (optional) one line: the name of the input directory |
| STATIC/        | results of analysis 101                              |
| DYNAMIC/       | results of analysis 103                              |
| WORK/          | matrix dumps and solver information                  |
| REPORT/        | WARNING_file.dat                                     |

# 1.6 The web interface

The folder `INTERFACE` contains a graphical **input editor and launcher**. Start it with `INTERFACE\Start-MUL2-Interface.cmd` and open `http://localhost:8765/` in a browser. It offers:

- **Editor:** one form per input file with checks of the cross-references between files;
- **Side-by-side** and **Raw text:** edit the text directly with the formatted view next to it;
- **3D Model:** a three-dimensional view of the structure (beams with their section, plates with exaggerated thickness, solids), supports and loads;
- **Run & Results:** send the current input set to the solver; read its console output and download the result files.

Use **New** to start from a template, **Load** to read files, **Folder** to work directly on a folder of your disk, **Save** and **Export** to write the files. Everything the interface does is writing input files and running the executable: you can always do the same by hand.

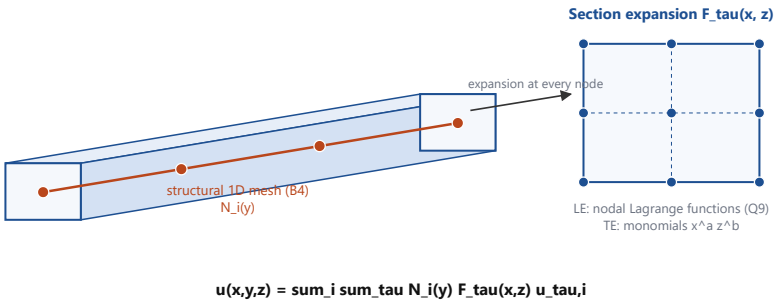
# How to describe a structure

## 2.1 Two meshes, not one

The unusual thing about a CUF model is that **a structure is described by two meshes**:

1. the **structural mesh** (`NODES.dat`, `CONNECTIVITY.dat`): what an ordinary finite element model has — nodes in space and elements joining them. A *beam* is a chain of line elements (B2, B3, B4) along its axis; a *plate* is a surface mesh (Q4, Q9, ...); a *solid* is a volume mesh (H8, H27, ...);
2. the **expansion mesh** (`EXP_MESH_nn.dat`, `EXP_CONN_nn.dat`): a small mesh of **what the structural element does not describe** — the cross-section of a beam, the thickness of a plate. Its nodes are local coordinates measured in the element frame, and each of its elements carries a *lamination* (a material and an orientation).

Each structural node also has an **expansion model** (`TEn` or `LE`): it decides how the displacement varies over the expansion mesh. With `TEn` the displacement of the section is a polynomial of degree  $n$ ; with `LE` it is interpolated from the displacements of the nodes of the expansion mesh.



**Figure 2.1.** A beam: structural nodes (left, red) and the section expansion mesh (right).

## 2.2 What the three families look like

**Table 2.1.** How the three families are modelled.

| Family | Structural elements | Expansion mesh                                  | Expansion nodes are at | Typical unknowns per node                                                                   |
|--------|---------------------|-------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------|
| Beam   | B2, B3, B4          | the cross-section: Q4, Q9, Q16, T3, T6 elements | $(x_s, 0, z_s)$        | $3T, T = \text{terms (TE n : } (n + 1)(n + 2) / 2; \text{ LE : nodes of the section mesh)}$ |
| Plate  | Q4, Q9, Q16, T3, T6 | the thickness: B2, B3, B4 elements              | $(0, 0, z_s)$          | $3T, T = \text{terms (TE n : } n + 1; \text{ LE : nodes of the thickness mesh)}$            |
| Solid  | H8, H27, ...        | one point (S1)                                  | $(0, 0, 0)$            | 3                                                                                           |

For a **solid** the section mesh is trivial: a single node and the element `S1` :

|                     |                 |
|---------------------|-----------------|
| EXP_MESH_01.dat     | EXP_CONN_01.dat |
| 1                   | 1               |
| 1 0.0D0 0.0D0 0.0D0 | S1 1 1 1        |

The expansion mesh is *shared* by all the structural elements that refer to it through their last integer in `CONNECTIVITY.dat` (the "section" number). A model can use several expansion meshes (`EXP_MESH_01`, `EXP_MESH_02`, ...); this is how a model can contain, for example, a thick-walled and a thin-walled part.

## 2.3 Units and coordinates

The program is **unit-free**: use any consistent set. The examples of this guide use SI units (metres, newtons, pascals, kilograms; frequencies in hertz).

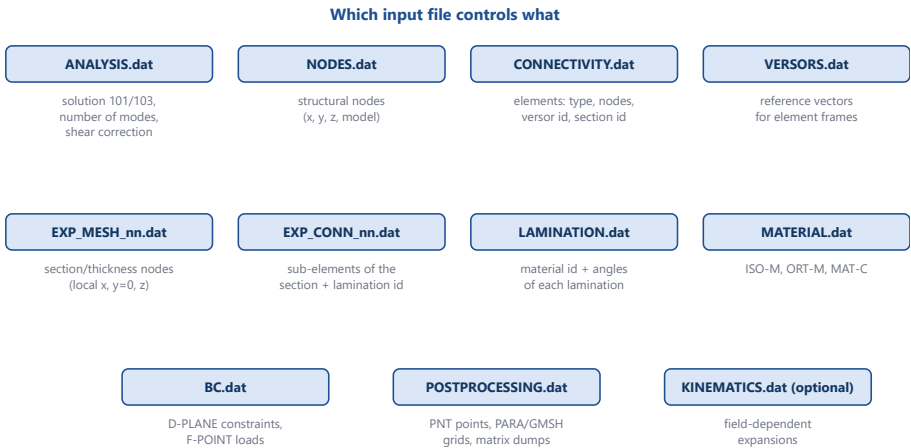
Three coordinate systems appear in the files (Chapter 5 explains them in detail):

- the **global system** ( $X, Y, Z$ ): node coordinates, point coordinates of loads and results, constraint planes;
- the **element system** ( $x, y, z$ ): used for the *section/thickness coordinates* in `EXP_MESH_nn.dat` and for strains and stresses in the `LOC` output; it is built automatically from the nodes and the *versor* of the element;
- the **material system** ( $T, L, Z$ ): the principal axes of a material, related to the element system by the angles of `LAMINATION.dat`.

## 2.4 Which element for which structure?

- **Beams:** the axis of the beam is the line through the nodes. Use B3 or B4 elements for curved or highly loaded beams and B2 for simple cases (use MITC for B2/B3 to avoid shear locking).
- **Plates and shells:** use Q4 (simple) or Q9 (more accurate, recommended with MITC). A plate may be oriented anywhere in space (the element frame follows its plane), but each element should be **flat or nearly flat**: the frame is built from three corners and out-of-plane warping of the nodes is neglected.
- **Solids:** H8 for simple blocks (without MITC), H27 for accuracy.
- **Mixed models:** a model may contain beams, plates and solids at the same time; they must be connected through *shared nodes* with compatible expansions (Chapter 6).

## 2.5 The files of a model



**Figure 2.2.** Which file controls what.

A minimal model needs: ANALYSIS.dat , NODES.dat , CONNECTIVITY.dat , VERSORS.dat , MATERIAL.dat , LAMINATION.dat , at least one EXP\_MESH\_01.dat + EXP\_CONN\_01.dat , BC.dat and POSTPROCESSING.dat . Chapter 4 describes each of them record by record.

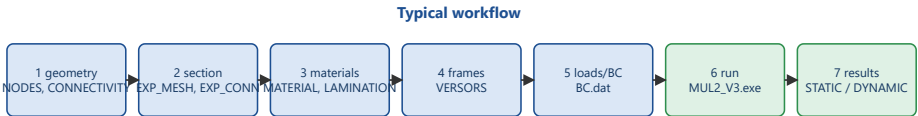
## 2.6 Rules that apply to all files

1. The first data line of every file is the **number of records** that follow.
2. Blank lines are ignored and so is everything that follows the data (descriptions, comments); the number of records tells where the data ends.
3. Fields are separated by blanks or tabs. Real numbers may be written as `1.0`, `1e-3` or `1.0D-3`.
4. Names (`B4`, `LE`, `ISO-M`, `D-PLANE`, ...) are not case sensitive.
5. Identifiers (nodes, elements, materials, ...) are positive integers and must be unique inside their file. They need not be consecutive.
6. The order of the files in the directory does not matter; all the files must be in the same input directory.

# Tutorial: a cantilever beam, step by step

In this chapter we build the input of a cantilever beam from scratch, run it, check the answer against beam theory and then change the model. All files are in `DOC/BOOKS/examples/cases/beam_te2_static/INPUT` and can be regenerated with `python DOC/BOOKS/examples/make_examples.py`.

**The problem.** A beam of length  $L = 1$  m, square section  $0.1 \times 0.1$  m, aluminium ( $E = 70$  GPa,  $\nu = 0.3$ ,  $\rho = 2700$  kg/m<sup>3</sup>). It is clamped at  $y = 0$  and a force  $P = 1000$  N along  $+z$  is applied at the centre of the free end. The beam axis is the global  $Y$  axis.



**Figure 3.1.** Workflow of the tutorial.

## 3.1 Step 1 – the analysis (ANALYSIS.dat)

```

101

4 modes

MITC
MITC
MITC
  
```

Record 1: the analysis (101 = static). Record 2 is the number of modes (used only by 103, ignored here). Records 3–5 select the shear-locking treatment for beams, plates and solids. We use `MITC` for all; it only acts on the family that is present.

## 3.2 Step 2 – the structural mesh (NODES.dat and CONNECTIVITY.dat)

We use five cubic beam elements (B4, four nodes each), so the beam has  $3 \times 5 + 1 = 16$  nodes spaced equally along  $Y$ , and every node uses a Taylor expansion of order 2 (TE 2):

NODES.dat (first records)

```
16

1  0.0D0  0.0000000000D+00  0.0D0  TE 2
2  0.0D0  6.6666666667D-02  0.0D0  TE 2
3  0.0D0  1.3333333333D-01  0.0D0  TE 2
4  0.0D0  2.0000000000D-01  0.0D0  TE 2
5  0.0D0  2.6666666667D-01  0.0D0  TE 2
6  0.0D0  3.3333333333D-01  0.0D0  TE 2
...  (10 more lines)
```

Each record is: node id,  $X$ ,  $Y$ ,  $Z$ , expansion model and its order. The elements join the nodes four by four:

CONNECTIVITY.dat

```
5

B4  1  1  2  3  4  1  1
B4  2  4  5  6  7  1  1
B4  3  7  8  9  10  1  1
B4  4  10 11 12 13  1  1
B4  5  13 14 15 16  1  1
```

A record is: element type, element id, the four node ids (in order along the axis), the **vector** id (1, see Step 4) and the **section** number (1 → EXP\_MESH\_01.dat).

## 3.3 Step 3 – the section (EXP\_MESH\_01.dat and EXP\_CONN\_01.dat)

The section is a single Q9 element,  $0.1 \times 0.1$  m, whose nine nodes sit at the section coordinates  $(x, 0, z)$ :

EXP\_MESH\_01.dat

```

9
1  -5.0000000000D-02  0.0D0  -5.0000000000D-02
2  0.0000000000D+00  0.0D0  -5.0000000000D-02
3  5.0000000000D-02  0.0D0  -5.0000000000D-02
4  -5.0000000000D-02  0.0D0  0.0000000000D+00
5  0.0000000000D+00  0.0D0  0.0000000000D+00
6  5.0000000000D-02  0.0D0  0.0000000000D+00
7  -5.0000000000D-02  0.0D0  5.0000000000D-02
8  0.0000000000D+00  0.0D0  5.0000000000D-02
9  5.0000000000D-02  0.0D0  5.0000000000D-02

```

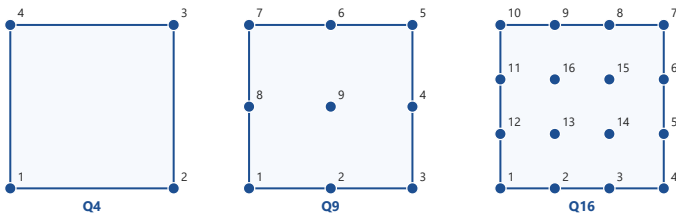
EXP\_CONN\_01.dat

```

1
Q9 1 1 1 2 3 6 9 8 7 4 5

```

The Q9 record reads: type, id, **lamination id** ( 1 ), and the nine node ids in the order of Figure 3.2 (corners and mid-sides alternate counter-clockwise, the centre last). With a Taylor expansion this mesh only supplies the integration domain and the places where loads and constraints can be applied (the nodes); the displacement is a polynomial of the section coordinates, not an interpolation between these nodes.



**Figure 3.2.** Node order of the quadrilateral elements.

### 3.4 Step 4 – the frame (VERSORS.dat)

```
1
VERSOR 1 0 0 1
```

A beam needs one extra piece of information: which way is "up" in its section. The versor (0, 0, 1) means that the local  $z$  axis of the section points along the global  $Z$ . The beam axis is the local  $y$  (here the global  $Y$ ) and local  $x$  completes the right-handed triad:  $x = y \times z = (1, 0, 0)$ . So the section coordinate  $x$  of EXP\_MESH\_01.dat runs along the global  $X$  and  $z$  along the global  $Z$ . (Chapter 5 shows how to choose the versor for other layouts.)

### 3.5 Step 5 – material and lamination (MATERIAL.dat, LAMINATION.dat)

```
1 1
ISO-M 1 7.00000000000D+10 0.3 2700.0
```

The first line says: one material, one record. ISO-M id E nu rho. And

```
1
LAM2 1 1 0.0 0.0
```

A *lamination* is a material (1) with two orientation angles (here both zero: nothing to rotate). The lamination id 1 is the one referred to by the section element.

### 3.6 Step 6 – supports and loads (BC.dat)

BC.dat

```
2
D-PLANE 1 0 1 0 0 0.0 0.0 0.0
F-POINT 2 0.0 1.00000000000D+00 0.0 0.0 0.0 1.00000000000D+03
```

D-PLANE 1 0 1 0 0 0.0 0.0 0.0 clamps every degree of freedom lying on the **plane**  $0 \cdot X + 1 \cdot Y + 0 \cdot Z + 0 = 0$ , that is  $Y = 0$ , to the three displacements 0.0 0.0 0.0 ( $u_x, u_y, u_z$ ). F-POINT 2 0 1 0 0 0 1000 is a force (0, 0, 1000) N at the point (0, 1, 0)

: the centre node of the section of the last node of the beam.

### WARNING

A point load must be applied **at a node of the section mesh** (here the centre node of the Q9), at a structural node. For a Taylor model the load is distributed over the polynomial terms, so the same  $P$  gives a different generalised force at a corner node than at the centre.

## 3.7 Step 7 – what to compute (POSTPROCESSING.dat)

POSTPROCESSING.dat

```
3

PARA 20 GLB 1 1 1 1 1 1 1 1 1
PNT 1 0.0 9.9999900000D-01 0.0
PNT 2 0.0 0.5 0.05
```

The first line says three requests follow. `PARA 20 GLB 1 1 1 1 1 1 1 1 1` writes a ParaView file with 20-node cells, results in the global frame (GLB), one cell per element (the nine 1 are subdivisions). `PNT 1 ...` and `PNT 2 ...` ask for the values at two points: just before the tip and at mid-length near the top face.

## 3.8 Step 8 – run

```
mkdir run & cd run
mkdir STATIC DYNAMIC WORK REPORT
xcopy /E /I ../INPUT INPUT
MUL2_V3.exe INPUT
```

## 3.9 Step 9 – read the results

`STATIC/POST_POINT.dat` has one row per point. The first row gives the tip displacement  $U_Z = 5.674\text{E-}04$  m. Beam theory predicts

$$\delta_{EB} = \frac{PL^3}{3EI} = 5.714 \times 10^{-4} \text{ m}, \quad \delta_{Timoshenko} = 5.759 \times 10^{-4} \text{ m},$$

so TE2 is 1.5 % lower than Timoshenko. The small difference is expected: the clamp of the model is a true three-dimensional clamp (the section cannot contract at  $Y = 0$ ), which stiffens the beam slightly.

Open `STATIC/RESULTS_PARA_01.vtk` in ParaView: choose `Displacements` as the "warp by vector" field to see the deformed beam, or colour by `Sigma_YY` to see the bending stress.

### 3.10 Step 10 – change the model

**Higher order.** Replace `TE 2` by `TE 4` in all the node records (the DOF grow from 288 to 720) and rerun: the tip deflection becomes  $5.685 \times 10^{-4}$  m. With `TE 1` it is  $4.282 \times 10^{-4}$  m (–26 %): *Poisson locking* (Chapter 4 of the Theoretical Guide). `TE2` is the lowest order for engineering use.

**Lagrange expansion.** Replace `TE 2` by `LE 2` (the second number is ignored for `LE`) and replace the section by a mesh of  $2 \times 2$  `Q9` elements (25 nodes). The tip deflection is  $5.689 \times 10^{-4}$  m with 1 200 DOF.

**Natural frequencies.** Change `ANALYSIS.dat` to `103` with 4 modes and remove the load from `BC.dat` (keep the clamp). `DYNAMIC/FREQUENCIES.dat` then contains

```
Frequency 1: 8.242627792E+001
Frequency 2: 8.242627792E+001
Frequency 3: 4.963753952E+002
Frequency 4: 4.963753952E+002
```

Modes 1 and 2 are the two bending modes (in the  $x$ – $y$  and in the  $y$ – $z$  plane: the same frequency because the section is square), modes 3 and 4 the second bending modes. Euler–Bernoulli gives  $f_1 = 82.25$  Hz and  $f_2 = 515.5$  Hz.

**Table 3.1.** The same beam with different models (5 B4 elements).

| Model                                 | DOF   | $w_{tip}$ [m]                                   | $f_1$ [Hz] |
|---------------------------------------|-------|-------------------------------------------------|------------|
| Theory (Euler–Bernoulli / Timoshenko) | –     | $5.714 \times 10^{-4}$ / $5.759 \times 10^{-4}$ | 82.25      |
| TE1                                   | 144   | $4.282 \times 10^{-4}$                          | 94.61      |
| TE2                                   | 288   | $5.674 \times 10^{-4}$                          | 82.43      |
| TE4                                   | 720   | $5.685 \times 10^{-4}$                          | 82.32      |
| LE (2×2 Q9)                           | 1 200 | $5.689 \times 10^{-4}$                          | 82.33      |
| Plate strip (Q9)                      | 459   | $5.666 \times 10^{-4}$                          | 82.48      |
| Solid (H27)                           | 459   | $5.663 \times 10^{-4}$                          | 82.53      |

### 3.11 Common mistakes in this tutorial

- **Forgetting the output directories** (`STATIC`, `DYNAMIC`, `WORK`, `REPORT`): the program creates them, but if you run it from a read-only folder it cannot.
- **A load that is not on a section node:** `POINT LOAD LOCATION WAS NOT FOUND`.
- **A section mesh with the wrong node order** in `EXP_CONN`: distorted or negative areas; the node order of the Q9 must follow Figure [Figure 3.2](#).
- **No constraint:** PARDISO error -4 (zero pivot) for a static problem, or a failed factorisation for the modal one.

# Input file reference

All files are in the input directory. For each file this chapter gives the purpose, the exact record syntax, the rules and the typical mistakes. In the syntax lines, [ ] marks an optional field and ... a repetition.

## 4.1 ANALYSIS.dat

Chooses the analysis and the numerical options.

|      |                        |                                                  |
|------|------------------------|--------------------------------------------------|
| 101  | STATIC                 | ← record 1: analysis number (+ optional comment) |
| 20   | Number of Modes ...    | ← record 2: number of modes (analysis 103)       |
| MITC | shear correction beam  | ← record 3: NONE   REDI   SELI   MITC            |
| MITC | shear correction plate | ← record 4                                       |
| NONE | shear correction solid | ← record 5                                       |

The historical files have an extra record between the analysis number and the number of modes (the old solver selector; e.g. 100 !! NOT USED ). It no longer exists: the program recognises it (a second integer line) and ignores it with a warning, so old inputs still run.

**Table 4.1.** Records of ANALYSIS.dat .

| Record       | Content                                                | Allowed values                                                                                                                                                         |
|--------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | analysis number                                        | 101 linear static; 103 free vibration. Other numbers of the historical program (104, 105, 106, 111, ...) give the error SOLUTION n IS NOT SUPPORTED (ONLY 101 AND 103) |
| 2            | number of modes                                        | integer ≥ 1; if larger than the number of free DOF a warning is issued and all are returned                                                                            |
| 3, 4, 5      | shear-locking treatment of beam, plate, solid elements | NONE full integration; REDI reduced integration; SELI selective integration (shear terms reduced, the others full); MITC tied shear strains; anything else is an error |
| 6 (optional) | physics that are solved                                | FIELDS MECH [THERM0] [PIEZ0] ; without the record every field expanded in KINEMATICS.dat is solved (Chapter 11)                                                        |

| Record          | Content                | Allowed values                                                                                                       |
|-----------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| 7<br>(optional) | joining by coincidence | JOIN COINCIDENT [tol] : the DOFs of different nodes that are the same field at the same point are joined (Chapter 6) |

### NOTE

MITC does nothing for topologies that have no tying table (Q16, T3, T6, T4, T10, P6): they are integrated fully and the program prints a warning. REDI and SELI reduce the Gauss rule of the *structural* element by one point per direction (B2 1, B3 2, B4 3 points; Q4 1, Q9 2, Q16 3; H8 1, H27 2); triangles stay fully integrated (warning). REDI leaves hourglass modes in Q4, H8 and H27 (warning): prefer SELI or MITC there. In a modal analysis REDI also reduces the mass.

## 4.2 NODES.dat – the structural nodes

Two formats are accepted.

**Historical format (default).** Each record: node id, three global coordinates, the expansion model and, for TE, its order:

|     |                   |                      |       |      |
|-----|-------------------|----------------------|-------|------|
| 16  | ← number of nodes |                      |       |      |
| 1   | 0.0D0             | 0.0000000000000000D0 | 0.0D0 | TE 2 |
| 2   | 0.0D0             | 0.0333333333333333D0 | 0.0D0 | TE 2 |
| ... |                   |                      |       |      |

**Table 4.2.** Historical NODES.dat records.

| Field  | Meaning                                                                          |
|--------|----------------------------------------------------------------------------------|
| ID     | positive integer, unique                                                         |
| X Y Z  | global coordinates                                                               |
| TE n   | Taylor expansion of order $n \geq 0$ for $u$ , $v$ and $w$                       |
| LE [m] | Lagrange expansion defined by the expansion mesh; the number is read and ignored |

Nodes that belong to no element are accepted but receive no unknowns (a warning is issued).

With `KINEMATICS.dat`. If the file `KINEMATICS.dat` exists the nodes carry a *kinematic id* instead of the model token:

```
KINEMATICS.dat
3

KINEMATIC 1  TE2 TE2 TE2 NONE NONE NONE NONE NONE NONE
KINEMATIC 2  LE  LE  LE  NONE NONE NONE NONE NONE NONE
KINEMATIC 3  TE4 TE2 TE2 NONE NONE NONE NONE NONE NONE

NODES.dat  (ID X Y Z KINEMATIC_ID)
3

1  0.0 0.0 0.0 1
2  0.0 0.1 0.0 2
3  0.0 0.2 0.0 3
```

The nine columns are the fields  $u, v, w, T, P, B, \sigma_{zz}, \sigma_{xz}, \sigma_{yz}$ . For analyses 101/103 only the first three may be different from `NONE`. Tokens: `TE $n$`  (Taylor order  $n$ ), `LE`, `NONE`; `HLE` (no order; the order belongs to the section sub-elements, Chapter 10) is supported; `Mn` is accepted syntactically but gives an explicit *not implemented* error. Different expansions for  $u, v$  and  $w$  at the same node are allowed.

**WARNING**

Two elements that share a node must use compatible expansions at that node. All the elements incident to a **Lagrange node** must use the **same expansion mesh** (error `LE NODE USES INCOMPATIBLE EXPANSIONS`); for **Taylor nodes** the expansion meshes of the incident elements must have the same dimension.

4.3 **CONNECTIVITY.dat – the structural elements**

```
5                                     ← number of elements

B4  1   1  2  3  4   1   1
B4  2   4  5  6  7   1   1
```

TYPE id node\_1 ... node\_N [vectors] section

Table 4.3. Element records.

| Field | Meaning                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| TYPE  | B2 B3 B4 (beams), Q4 Q9 Q16 T3 T6 (plates; Q3, Q6 = T3, T6), H8 H27 (solids). H20, T4, T10, P6 are recognised but not implemented (error) |

| Field               | Meaning                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| id                  | positive integer, unique                                                                                           |
| node_1...<br>node_N | the $N$ node ids in the element node order (Figure <a href="#">Figure 3.2</a> and the quick reference, Appendix A) |
| versor              | id of the versor in <code>VERSORS.dat</code> (beams and plates); for solids it may be omitted or any value         |
| section             | number $nn$ of the expansion mesh <code>EXP_MESH_nn.dat</code> / <code>EXP_CONN_nn.dat</code> used by the element  |

If only one integer follows the nodes it is taken as the **section** number and the versor as 0.

Rules: node ids must exist; at most 99 expansion meshes; the section numbers used must be 1 ...  $n_{max}$  without gaps (every file `EXP_*_01 ... nn` for  $nn \leq n_{max}$  must exist).

#### 4.4 VERSORS.dat – the reference vectors

```

2

VERSOR 1  0 0 1
VERSOR 2  1 0 0

```

`VERSOR id vx vy vz`. The vector fixes the rotation of the **element frame**: for a beam it gives the direction of the local  $z$  axis (the "up" of the section), for a plate the direction of the local  $x$  axis in the plane of the element. It need not be normalised and, for a beam, may be any vector that is not parallel to the axis; for a plate any vector that is not normal to the element. Chapter 5 gives recipes. A zero vector is an error.

#### 4.5 MATERIAL.dat – materials

```

2 2                                     ← number of materials, number of records

ISO-M  1  7.0D10  0.30  2700.0
ORT-M  2  140D9 10D9 10D9  0.3 0.3 0.4  5D9 5D9 3.5D9  1600

```

The header has two integers: the number of **mechanical materials** and the number of **records** that follow (the file may contain additional, non-mechanical records of the historical program, which are accepted and ignored).

**Table 4.4.** Material records.

| Record                                                                                                    | Fields                                           | Notes                                                                                                           |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| ISO-M                                                                                                     | id E nu rho                                      | isotropic; $E > 0$ , $-1 < \nu < 0.5$ , $\rho \geq 0$                                                           |
| ORT-M                                                                                                     | id EL ET EZ<br>nuLT nuLZ nuTZ<br>GLT GLZ GTZ rho | orthotropic in the axes $(T, L, Z)$ : $L$ fibre, $T$ transverse in the plane, $Z$ thickness                     |
| MAT-C                                                                                                     | id and 36<br>numbers (row by<br>row) and rho     | full $6 \times 6$ stiffness in the material axes, Voigt order $(TT, LL, ZZ, TZ, LZ, TL)$ ; it must be symmetric |
| T-EXP S-EXP T-SPC VISCO T-<br>CON Z-EXP Z-PRM M-EXP M-PRM<br>H-EXP H-DIF PIROE PIROM PZ-<br>MG MZ-PZ DAMP |                                                  | accepted and ignored in 101/103                                                                                 |

**NOTE**

In an orthotropic material the **fibre direction is  $L$** . With zero lamination angles  $T \equiv x$ ,  $L \equiv y$ ,  $Z \equiv z$  of the element frame: for a beam, fibres run along the beam axis; for a plate, fibres run along the element's  $y$  direction.

**4.6 LAMINATION.dat – material orientation**

```
3
LAM2 1 1 0.0 0.0
LAM2 2 1 0.0 90.0
LAM2 3 1 0.0 0.0
```

LAM2 id material\_id angle\_y angle\_z (degrees). A lamination is a material with an orientation: the angle  $\theta_y$  rotates the material axes about the element  $y$  axis and  $\theta_z$  about the element  $z$  axis (Chapter 5). The same material may appear in several laminations with different angles; this is how a laminate is described: one lamination per ply (or per group of plies), each referenced by a sub-element of the expansion mesh. Only LAM2 records are accepted in 101/103.

# 4.7 EXP\_MESH\_nn.dat and EXP\_CONN\_nn.dat – the section or thickness

Each expansion mesh is a pair of files, with nn = 01, 02, ....

|                         |                          |
|-------------------------|--------------------------|
| EXP_MESH_01.dat         | EXP_CONN_01.dat          |
| 9                       | 1                        |
| 1 -0.05D0 0.0D0 -0.05D0 | Q9 1 1 1 2 3 6 9 8 7 4 5 |
| 2 0.00D0 0.0D0 -0.05D0  |                          |
| ...                     |                          |
| 9 0.05D0 0.0D0 0.05D0   |                          |

**EXP\_MESH\_nn.dat:** id x y z — the **local** coordinates of the expansion nodes, measured in the element frame from the structural node. For a **beam** the section lies in the plane  $y = 0$ : x, z in the section,  $y = 0$ . For a **plate** the thickness is along z:  $x = y = 0$ . For a **solid** a single node at the origin.

**EXP\_CONN\_nn.dat:** TYPE id lamination node\_1 ... node\_N — the sub-elements. Allowed types: for beams Q4 Q9 Q16 T3 T6; for plates B2 B3 B4; for solids S1. The node ids refer to EXP\_MESH\_nn.dat. The **lamination id** says which row of LAMINATION.dat (material and angles) the sub-element has.

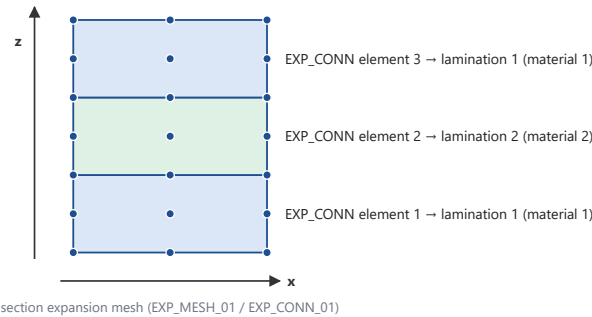


Figure 4.1. A laminated section: one sub-element per ply.

Rules:

- The mesh must be **non-degenerate**: the nodes must span the two section axes (beam) or the thickness axis (plate).
- Sub-elements must cover the section without overlapping; nodes shared by two sub-elements must be the same node.

- For a Taylor expansion the expansion mesh defines the integration domain and the points where loads and constraints can be placed.
- The node order of every sub-element follows its topology (Figure [Figure 3.2](#) and the quick reference, Appendix A).

## 4.8 BC.dat – constraints and loads

```

2

D-PLANE 1 0 1 0 0 0.0 0.0 0.0
F-POINT 2 0.0 1.0 0.0 0.0 0.0 1000.0

```

D-PLANE id A B C D ux uy uz — prescribes displacements on every degree of freedom that lies on the plane

$$A X + B Y + C Z + D = 0 .$$

ux uy uz are the prescribed values of the global displacement components; write N for a component that is left free. Examples:

**Table 4.5.** Examples of D-PLANE .

| Record                         | Effect                                            |
|--------------------------------|---------------------------------------------------|
| D-PLANE 1 0 1 0 0 0.0 0.0 0.0  | plane $Y = 0$ : all three components clamped to 0 |
| D-PLANE 2 0 1 0 -1 N 0.0 N     | plane $Y = 1$ ( $Y - 1 = 0$ ): only $u_y$ fixed   |
| D-PLANE 3 0 0 1 0 N N 0.0      | plane $Z = 0$ : $u_z = 0$ (symmetry plane)        |
| D-PLANE 4 1 1 0 -2 0.0 0.0 0.0 | oblique plane $X + Y = 2$                         |

The test is done on the **physical position of every expansion node of every structural node**, within a small tolerance. For a Lagrange expansion only the section nodes that lie on the plane are constrained (a plane may cut the section: partial constraints are possible). For a **Taylor expansion the plane must contain the whole section** of the node: the whole expansion is then constrained (constant term to the value, the other terms to zero); a plane that cuts a Taylor section gives the error `PARTIAL PLANE CONSTRAINT REQUIRES MPC FOR TE`.

The plane equation is  $AX + BY + CZ + D = 0$  — note the **plus**  $D$ . The plane  $Y = 1$  is  $0 \ 1 \ 0 \ -1$ , not  $0 \ 1 \ 0 \ 1$ . A plane that selects no degree of freedom gives the warning `PLANE DID NOT SELECT ANY ACTIVE DOF`; two records that prescribe different values for the same DOF give an error.

In **analysis 103** loads and non-zero prescribed values are ignored (warning); only the *constraints* matter.

| Record          | Meaning                                                                        |
|-----------------|--------------------------------------------------------------------------------|
| PNT id x<br>y z | write the results at the global point $(x, y, z)$ in POST_POINT.dat            |
| KMAT ,<br>MMAT  | write the stiffness / mass matrix in WORK/K_MAT.dat , WORK/M_MAT.dat           |
| FRCE ,<br>UNKN  | write the force vector / solution vector in WORK/FORCES.dat , WORK/UNKNOWN.dat |
| ENRG            | accepted, no output in this release                                            |

The subdivision numbers refine the output grid *inside every element and sub-element* (use 2 or 3 to see smooth curves along the beam axis; each sub-element of the section is a separate set of cells so that laminate layers are visible). The meaning of  $(\xi, \eta, \nu)$  is: for a beam  $\eta$  along the axis,  $(\xi, \nu)$  over the section; for a plate  $(\xi, \eta)$  in the surface and  $\nu$  through the thickness; for a solid the three element coordinates.

If the declared number of records is larger than the number present, the missing ones are ignored with a warning. Unknown record words are ignored with a warning. A point outside the structure gives a row of NaN in POST\_POINT.dat and a warning.

## 4.10 A complete minimal checklist

**Table 4.7.** Checklist before running.

| Check                                                                                             | Where                                    |
|---------------------------------------------------------------------------------------------------|------------------------------------------|
| analysis is 101 or 103; modes requested for 103                                                   | ANALYSIS.dat                             |
| every element refers to existing nodes, an existing versor and a section number whose files exist | CONNECTIVITY.dat , VERSORS.dat , EXP_*   |
| node order of every element follows the figures                                                   | CONNECTIVITY.dat , EXP_CONN_nn.dat       |
| the laminations used by the sub-elements exist; each lamination refers to an existing material    | EXP_CONN , LAMINATION.dat , MATERIAL.dat |
| the structure is supported: no rigid-body motion left                                             | BC.dat                                   |
| loads are at section nodes (static only)                                                          | BC.dat                                   |
| at least one PNT or PARA request                                                                  | POSTPROCESSING.dat                       |

# Reference systems in practice

This chapter explains, with recipes and examples, how to orient beams, plates and materials. The theory is in Chapter 5 of the Theoretical Guide; here the question is: *given my structure, what do I write in the files?*

## 5.1 The three systems at a glance

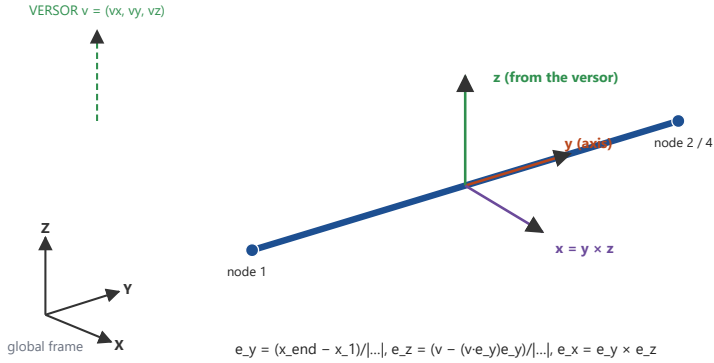
**Table 5.1.** The three reference systems.

| System          | Symbol    | Defined by                                             | Used for                                                                                         |
|-----------------|-----------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Global          | $X, Y, Z$ | the coordinates in <code>NODES.dat</code>              | node coordinates, loads, constraint planes, displacements in the output, <code>GLB</code> output |
| Element (local) | $x, y, z$ | element geometry + <code>VERSORS.dat</code>            | the coordinates of <code>EXP_MESH_nn.dat</code> ; strain and stress in <code>LOC</code> output   |
| Material        | $T, L, Z$ | element system + angles of <code>LAMINATION.dat</code> | the constants of <code>ORT-M</code> and <code>MAT-C</code>                                       |

The three axes of the element system are always right-handed.

## 5.2 Beams

For a beam element the local  $y$  axis **is the axis of the beam**, from its first node to its last node. The versor  $v$  gives the **direction of the local  $z$  axis** of the section (the part of  $v$  along the axis is ignored), and the local  $x$  is  $y \times z$ .



**Figure 5.1.** Beam frame.

The section coordinates written in `EXP_MESH_nn.dat` are measured on these axes: a node with  $x = 0.05$ ,  $z = -0.05$  lies 0.05 along the local  $x$  and 0.05 against the local  $z$  from the axis. The  $y$  coordinate of a section node is always 0.

**Table 5.2.** Beam recipes (axis direction, versor  $\rightarrow$  local axes in global components).

| Beam axis (first $\rightarrow$ last node) | Versor    | Local $x$ | Local $y$ | Local $z$ |
|-------------------------------------------|-----------|-----------|-----------|-----------|
| +Y                                        | (0, 0, 1) | +X        | +Y        | +Z        |
| +Y                                        | (1, 0, 0) | -Z        | +Y        | +X        |
| +X                                        | (0, 0, 1) | -Y        | +X        | +Z        |
| +X                                        | (0, 1, 0) | +Z        | +X        | +Y        |
| +Z                                        | (1, 0, 0) | +Y        | +Z        | +X        |
| +Z                                        | (0, 1, 0) | -X        | +Z        | +Y        |

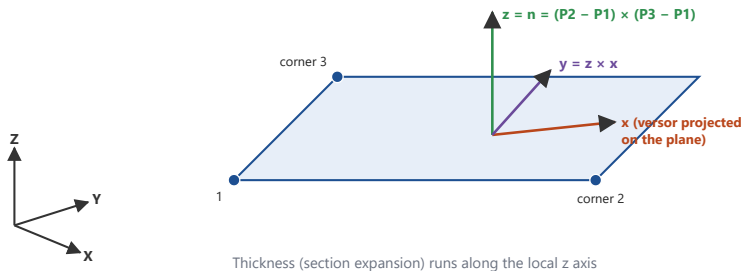
**Rule of thumb.** Choose the versor along the direction that you want the *height* of the section to run (the axis whose coordinate you will put in  $z$ ), and put the *width* in  $x$ . For a beam along  $Y$  with the section height along  $Z$  use (0, 0, 1).

**Sign matters.** Reversing the versor ( $v \rightarrow -v$ ) flips both  $z$  and  $x$  (a rotation by  $180^\circ$  of the section about the axis). It does not matter for a doubly-symmetric section; it does for an unsymmetric one.

**Beam elements of one beam must have the same orientation.** Elements that belong to the same straight beam can share the same versor id. A beam that bends (curved beam, frame) needs a versor per straight part: give each part its own `VERSOR` record and refer to it in the element records.

### 5.3 Plates

For a plate the element frame is built from three corner nodes of the element (Q4: nodes 1, 2, 4; Q9: 1, 3, 7; Q16: 1, 4, 10; triangles: 1, 2, 3). The local  $z$  axis is the **normal** (right-hand rule applied to node 1→2 and 1→3), the local  $x$  is the **projection of the versor on the plate**, and  $y = z \times x$ .



**Figure 5.2.** Plate frame.

The **thickness coordinate** written in `EXP_MESH_nn.dat` (third column) is measured along the local  $z$ : positive values are on the side the normal points to ("top").

**Table 5.3.** Plate recipes (the nodes of the element are numbered counter-clockwise looking from the side of the normal).

| Plate plane | Node order (corners)                                                | Normal $z$ | Versor      | Local $x$ | Local $y$ |
|-------------|---------------------------------------------------------------------|------------|-------------|-----------|-----------|
| $XY$        | counter-clockwise seen from $+Z$                                    | $+Z$       | $(1, 0, 0)$ | $+X$      | $+Y$      |
| $XY$        | clockwise seen from $+Z$                                            | $-Z$       | $(1, 0, 0)$ | $+X$      | $-Y$      |
| $XZ$        | $X$ then $Z$ direction (corner 2 along $+X$ , corner 3 along $+Z$ ) | $-Y$       | $(1, 0, 0)$ | $+X$      | $+Z$      |

| Plate plane | Node order (corners)                 | Normal $z$ | Versor    | Local $x$ | Local $y$ |
|-------------|--------------------------------------|------------|-----------|-----------|-----------|
| YZ          | corner 2 along +Y, corner 3 along +Z | +X         | (0, 1, 0) | +Y        | +Z        |

**Practical advice.** Number the nodes of every plate element in the same sense (counter-clockwise seen from the side you call "top") and use one versor per plane. With a laminate, "top" is also the side of the *last* ply of the thickness mesh: the order of the plies in `EXP_CONN` goes from the most negative to the most positive  $z$  (bottom to top).

## 5.4 Solids

For solid elements the element frame **is** the global frame. Nothing to choose. The versor field of the element record is ignored.

## 5.5 Mixing beams, plates and solids

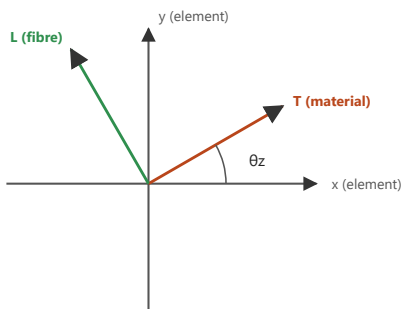
Elements of different families that meet at a node must use compatible expansions at that node. The structural coordinates  $(X, Y, Z)$  of the node are global; each element uses its own local frame, which is built independently. The unknowns are always global displacement components, so no transformation at the junction is necessary.

## 5.6 Material orientation

The elastic constants of `ORT-M` are given in the material axes  $(T, L, Z)$ . With zero angles they coincide with the element axes  $(x, y, z)$ :

- **beam:**  $L = y$  (the fibre runs along the beam axis),  $T = x, Z = z$ ;
- **plate:**  $L = y$  (along the second in-plane local axis),  $T = x, Z = z$  (through the thickness).

The two angles of a lamination, in degrees, rotate the material relative to the element system:



#### Material axes (T, L, Z)

LAMINATION.dat gives two angles per lamination  
 ANGLE\_Y: rotation about the element y axis  
 ANGLE\_Z: rotation about the element z axis  
 $A = R_z(\theta_z) \cdot R_y(\theta_y)$  maps element-frame compone to material-frame components.  
 With  $\theta_y = 0$  and  $\theta_z = 0$  the material axes  
 T, L, Z coincide with the element x, y, z.

**Figure 5.3.** Rotation of the material axes by  $\theta_z$  about the element z axis.

- $\theta_z$  (angle\_z): rotation about the local z. For a *plate*, this turns the fibres in the plane of the plate:  $\theta_z = 90^\circ$  puts the fibres along *x* instead of *y*. For a *beam* it turns the fibres in the plane of the section.
- $\theta_y$  (angle\_y): rotation about the local *y*, which for a plate means a rotation in the thickness plane and for a beam a rotation of the fibres about the axis.

#### NOTE

The matrix  $R_z(\theta_z)R_y(\theta_y)$  maps *element* components to *material* components; so a positive  $\theta_z$  turns the material axes counter-clockwise when looking from  $+z$  (as drawn above) — to check the sign in a new model, run a coupon with a known answer such as the example below.

### 5.6.1 Example: stacking sequence of a plate strip

A cantilever strip  $0.1 \times 1.0$  m with thickness 0.1 m of three equal plies, material ORT-M 1 140D9 10D9 10D9 0.3 0.3 0.4 5D9 5D9 3.5D9 1600 (fibre along *L*). The strip runs along *Y* and has the thickness along *Z*, so the fibre direction for zero angles is along the strip. The plies are given by three laminations:

**Table 5.4.** Two stacking sequences of the same strip (clamped at  $Y = 0$ , tip load 1000 N at the top face).

| Laminations (bottom → top)                | angle_z of the plies | Tip w [m]             | $f_1$ [Hz] |
|-------------------------------------------|----------------------|-----------------------|------------|
| LAM2 1 1 0 0, LAM2 2 1 0 90, LAM2 3 1 0 0 | 0°/90°/0°            | $3.22 \times 10^{-4}$ | 120.97     |

| Laminations (bottom → top)                 | angle_z of the plies | Tip $w$ [m]           | $f_1$ [Hz] |
|--------------------------------------------|----------------------|-----------------------|------------|
| LAM2 1 1 0 90, LAM2 2 1 0 0, LAM2 3 1 0 90 | 90°/0°/90°           | $2.67 \times 10^{-3}$ | 49.23      |

With 0°/90°/0° the two *outer* plies (which carry the bending) have their fibres along the strip, the strip is **8 times stiffer** than in the 90°/0°/90° case, where the stiff fibre ply is in the middle near the neutral axis and the outer plies are matrix-dominated. This is a good quick check of the angle convention in any new model: *outer plies with fibres along the beam axis ⇒ stiff*.

The files are in `examples/cases/plate_lam_0_90_0_static` and `plate_lam_90_0_90_static`; the expansion mesh is three B2 sub-elements:

EXP\_MESH\_01.dat (thickness nodes, bottom to top)

```

4
1  0.0D0 0.0D0 -5.0000000000D-02
2  0.0D0 0.0D0 -1.6666666667D-02
3  0.0D0 0.0D0 1.6666666667D-02
4  0.0D0 0.0D0 5.0000000000D-02

```

EXP\_CONN\_01.dat (one sub-element per ply, with its lamination id)

```

3
B2 1 1 1 2
B2 2 2 2 3
B2 3 3 3 4

```

LAMINATION.dat

```

3
LAM2 1 1 0.0 0.0000000000D+00
LAM2 2 1 0.0 9.0000000000D+01
LAM2 3 1 0.0 0.0000000000D+00

```

### 5.7 Output frames

Displacements are **always** written in the global frame in `POST_POINT.dat` (columns `u_x`, `u_y`, `u_z`). Strains and stresses are written in **both** frames: the (LOC) columns in the element frame of the element that contains the point, the (GLB) columns in

the global frame. In the VTK/GMSH files the `LOC` / `GLB` word of the request selects the frame of **all** the output fields (displacement, strain, stress):

- `GLB` : all components along  $X, Y, Z$ ;
- `LOC` : components along the element axes  $x, y, z$  of each element (for a beam  $y$  is the axis, so `Sigma_YY` is the axial stress; for a plate `Sigma_ZZ` is the through-thickness stress).

Strain and stress component order is always  $(xx, yy, zz, xz, yz, xy)$  with engineering shear strains  $\gamma$ .

# Modelling guide

This chapter collects advice on how to build good models: how to choose the elements and the expansion, how fine the meshes should be, how to support and load the structure, and how to check the answer:

## 6.1 Choosing the structural theory

Table 6.1. Decision guide.

| You want...                                                          | Use                                                   | Why                                                                |
|----------------------------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| global response (deflection, frequencies) of a slender beam          | beam, TE 2                                            | the cheapest model that includes the Poisson effect of the section |
| local effects, thick or short beam, non-compact sections             | beam, TE 3 – TE 6 , or LE                             | richer section kinematics                                          |
| thin-walled, open or hollow section, stress concentration at a joint | beam with an LE section mesh                          | the mesh follows the geometry; no ill-conditioning                 |
| laminated or sandwich beam or plate                                  | LE with one sub-element per ply or per group of plies | displacement continuity between plies, layer-wise stresses         |
| thin plate                                                           | plate Q9 (or Q4) with MITC , thickness TE 2 or LE     | locking-free                                                       |
| full 3D                                                              | H27 solid                                             | no kinematic assumption                                            |

**Order of cost** (per structural node): TE 1 (9 DOF) < TE 2 (18) < LE with a few section nodes < TE 4 (45) < ... The assembly and solution cost grow faster than the number of unknowns; a good strategy is to start with a coarse model and refine.

## 6.2 Mesh along the axis or in the plane

- Use **B4** (cubic) or **B3** elements along the beam when the loading is concentrated or the beam is highly loaded; at least **4 to 5 B3/B4 elements** per beam give a sound global response.

- Use **B2** only with `MITC`, and with 10 or more elements per beam.
- For plates use **Q9** with `MITC` and at least  $4 \times 4$  elements per panel; refine near supports, loads and holes.
- Remember that each structural node carries the *whole* section expansion: refining along the axis multiplies the number of unknowns by the number of terms.

## 6.3 Choosing the section mesh (Lagrange expansions)

- Use **Q9** sub-elements. One element per flat part of the cross-section gives a first answer;  $2 \times 2$  gives a robust one.
- For bending the **height** needs more elements than the width; for a thin-walled section place at least 2–3 elements across the thickness of every wall to capture bending of the wall.
- For a laminate, use **at least one sub-element per ply** through the thickness (B2 sub-elements for a plate give a linear distribution in each ply; B3 gives a quadratic one). More sub-elements per ply improve the transverse stresses.
- The mesh must be **conforming**: nodes shared by two sub-elements must be the same node.
- For a Taylor expansion any simple convex section mesh is fine (a single Q9 or Q4 for rectangular sections; several sub-elements for a section made of several laminations).

## 6.4 Coupling different parts of the model

All the elements incident to a node must be consistent with the **expansion model of that node**:

- A *Lagrange* node requires that **every** element sharing it refers to the **same expansion mesh** (same section number). A beam of section 1 and a beam of section 2 can therefore not be joined at a Lagrange node; use a Taylor model, or give both beams the same section mesh and orient them with versors.
- A *Taylor* node requires that all incident elements use expansion meshes of the same **dimension** (all beam sections or all plate thicknesses). Elements may have different section shapes but the Taylor order is that of the node.
- The model may contain beams, plates and solids at the same time (for example a beam, a plate and a block analysed together in separate regions), but a node

cannot be shared by elements of different families in this release.

In practice, to connect two different structural families, model the junction region with the more general family (for instance a short solid transition) or analyse the two parts separately with equivalent loads.

## 6.5 Joining by coincidence of the DOFs ( JOIN COINCIDENT )

The ordinary joint is the **shared node**: two elements that cite the same node number share its degrees of freedom. When two parts are modelled separately, their structural nodes do not coincide even if their expansion nodes do: a beam whose section node lies on a plate, a plate and a solid that touch along a face. For these cases the last record of ANALYSIS.dat can be

```
JOIN COINCIDENT
```

(optionally followed by a tolerance, a fraction of the diagonal of the model, default  $10^{-6}$ ). The program then identifies two degrees of freedom of different nodes when they are **the same field at the same point of space**, the point being the node plus the offset of the expansion node:

| Expansion | What is joined                                                                                                                                             |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LE        | every expansion node of a node with those of other nodes at the same point (same field)                                                                    |
| TE        | the term 1 (the value at the node); the higher terms only if the two elements have the same orientation, since they are derivatives along the element axes |
| HLE       | nothing (side and internal modes are not values)                                                                                                           |

The joined DOFs are one DOF of the system. The run prints a warning with the outcome (DOFs joined, or none found). Without the record nothing changes. Remarks: nodes that must stay separate at the same position (a crack) are joined too, so use the record only when you want the joint; an LE DOF is never joined to a TE one; shells with the rotation of the node at an edge are not joined by this record (use shared nodes).

Test: TESTS/JOIN/join\_tests.py (two beam segments whose structural nodes are shifted and whose section meshes are shifted back give the same result as the beam with a shared node, to the last digit).

## 6.6 Supports

- **Clamps:** a `D-PLANE` at the end section constrains *all* DOFs of that section (every term of every node on the plane). It is a *perfect three-dimensional clamp*: the section cannot warp or contract at the root, and the structure is slightly stiffer than the beam theory predicts.
- **Simple supports and symmetry:** constrain only the components that must vanish, writing `N` for the others (see the examples of `D-PLANE`), and remember that a **Taylor** node must lie completely on the plane.
- **Avoid rigid-body motions:** a static analysis needs at least the six rigid-body motions of each connected part to be constrained; otherwise PARDISO reports a zero pivot. A modal analysis with free rigid-body modes fails for the same reason.
- **Constrain only what is needed:** over-constraining (for example all three components at all the nodes of a long edge) creates artificial stiffness and local stress peaks.

## 6.7 Loads

- Only **point loads** exist. They must be applied at nodes of the section mesh of a structural node. To model a distributed load, divide the total force among the nodes consistently: for a uniform load on a Q9 edge use  $1/6$ ,  $4/6$ ,  $1/6$  of the total on the corner, mid and corner node; on a Q4 edge  $1/2$  and  $1/2$ ; on a B4 beam axis load  $q$  per unit length use  $q\ell$  ( $1/8$ ,  $3/8$ ,  $3/8$ ,  $1/8$ ) on the four nodes of an element of length  $\ell$  (each node shared between two elements receives the contribution of both).
- With a **Taylor expansion** a point load at a section node is *projected* on all the monomials: the same force at a corner and at the centre produces different generalised forces. A resultant at the centroid is best applied at the **centre node** of the section mesh.
- For a **tip shear on a section**, a single concentrated load produces local stress peaks at the load point (Saint-Venant): compare stresses at least one section height away from the point of application.

## 6.8 Mesh convergence and model checks

1. **Refine and compare.** Double the number of elements along the axis (and/or add section elements) and verify that the quantity you care about changes by less than your tolerance.
2. **Compare expansions.** A `TE 2` result that differs more than 2–3 % from `TE 4` or `LE` signals that the model is too poor for the problem.
3. **Use a simple theory as a cross-check:** deflection  $PL^3 / 3EI$ , frequencies of Euler–Bernoulli beams, etc. (Chapter 9).
4. **Look at the deformed shape** (ParaView *Warp by vector*) before reading numbers: it reveals missing supports, wrong node ordering and wrong units immediately.
5. **Read** `REPORT/WARNING_file.dat` and the console after every run.
6. **Check the equilibrium** of the reaction roughly: the sum of the loads must be balanced by the supports (use the dumps `KMAT` and `FRCE` if you need the reactions).

## 6.9 Strains and stresses

Strains and stresses are computed from the displacement field *at the point* and are **discontinuous** between elements and between plies. Do not average them blindly:

- At a point exactly on an element or sub-element boundary the program picks one of the two sides; ask for points slightly inside a ply (a small distance from the interface) to read the ply stresses.
- Transverse shear and normal stresses of laminates are good only with a fine thickness mesh (several sub-elements per ply) and quadratic Lagrange elements.
- Stresses near constraints, loads and re-entrant corners depend on the mesh size: they are *singular* in the continuum and converge slowly.

## 6.10 Numerical hygiene

- **Scale.** Work with dimensions of order one whenever possible. For high-order Taylor models measure the section in units where its size is about 1 (for example millimetres instead of metres), because monomials of order  $n$  of a very small number underflow.

- **Aspect ratios.** Elements with very large aspect ratios and thin plates with `NONE` give poorly conditioned matrices; use `MITC`.
- **Do not use `MITC` on H8 elements.**
- **High Taylor orders ( $> 8$ )** can make the matrices numerically singular. The program then retries with a more robust factorisation and prints a warning; check the answer with a different expansion.

### 6.11 Memory and time

**Table 6.3.** Rough cost of beam models (one run, Release build, 16 cores).

| Model                                              | DOF     | Non-zeros of K | Peak memory | Static time |
|----------------------------------------------------|---------|----------------|-------------|-------------|
| 5 B4, <code>TE 2</code>                            | 288     | 25 000         | < 50 MB     | < 0.3 s     |
| 5 B4, <code>LE (2 × 2 Q9)</code>                   | 1 200   | 0.2 M          | < 100 MB    | < 0.5 s     |
| 136 B4, 16 Q9 section elements ( <code>LE</code> ) | 99 387  | 20 M           | 1.4 GB      | 6 s         |
| 82 B4, <code>TE 15</code>                          | 100 776 | 205 M          | 7.8 GB      | 26 s        |

Estimate memory as: **16 bytes × (non-zeros of K)** for the matrix, **plus** the factor of PARDISO (typically 2–10 times the matrix for three-dimensional-like coupling), **plus** the same again for  $M$  in a modal analysis. Taylor models are dense inside each element ( $n^2$  non-zeros per element) while Lagrange models only couple section nodes that share a sub-element.

# Output reference

## 7.1 Where the results go

After a run the following folders exist in the run directory:

**Table 7.1.** Output files.

| File                                                     | When                                | Content                                      |
|----------------------------------------------------------|-------------------------------------|----------------------------------------------|
| STATIC/POST_POINT.dat                                    | 101, if PNT requested               | values at the requested points               |
| STATIC/RESULTS_PARA_nn.vtk                               | 101, one per PARA record            | fields on a grid, for ParaView               |
| STATIC/RESULTS_GMSH_nn.msh                               | 101, one per GMSH record            | the same for GMSH                            |
| DYNAMIC/FREQUENCIES.dat                                  | 103                                 | natural frequencies in Hz                    |
| DYNAMIC/RESULTS_DYN_PARA.vtk ,<br>RESULTS_DYN_GMSH.msh   | 103                                 | mode shapes                                  |
| WORK/K_MAT.dat , M_MAT.dat , FORCES.dat ,<br>UNKNOWN.dat | if KMAT MMAT FRCE<br>UNKN requested | matrices and vectors                         |
| WORK/ARPACK_SOLVER_INFO.dat                              | 103                                 | eigenvalues, residuals,<br>solver statistics |
| REPORT/WARNING_file.dat                                  | always                              | warnings of the run                          |

## 7.2 POST\_POINT.dat

A header line and one row per PNT request. Each row has 46 numbers. The columns are, in order:

**Table 7.2.** Columns of POST\_POINT.dat .

| Columns | Name                        | Meaning                                                                   |
|---------|-----------------------------|---------------------------------------------------------------------------|
| 1       | ID                          | point id of the request                                                   |
| 2–4     | X_EVAL , Y_EVAL ,<br>Z_EVAL | position where the field was <b>actually evaluated</b> (inside the model) |

| Columns | Name                       | Meaning                                                     |
|---------|----------------------------|-------------------------------------------------------------|
| 5–7     | X_REQ, Y_REQ, Z_REQ        | position requested                                          |
| 8       | LAM                        | id of the lamination at the point                           |
| 9       | ELE                        | element id that contains the point                          |
| 10      | S_ELE                      | id of the sub-element (expansion element)                   |
| 11–13   | U_X, U_Y, U_Z              | displacement, <b>global</b> components                      |
| 14–19   | EPS_XX ... EPS_XY<br>(LOC) | strain in the element frame, order $xx, yy, zz, xz, yz, xy$ |
| 20–25   | EPS_... (GLB)              | strain in the global frame                                  |
| 26–31   | SIG_XX ... SIG_XY<br>(LOC) | stress in the element frame                                 |
| 32–37   | SIG_... (GLB)              | stress in the global frame                                  |
| 38–40   | DT, VOLT, %H               | zeros (kept for compatibility)                              |
| 41–46   | E11 ... D33                | zeros (kept for compatibility)                              |

The shear strains are **engineering** shears:  $\gamma_{xz} = 2\epsilon_{xz}$ . The shear stresses are the usual  $\sigma_{xz}$  etc.

If a point lies outside the model, its row has `NaN` in all result columns and a warning is added to `REPORT/WARNING_file.dat`.

### NOTE

Reading a row in a script: skip the header, split the line on blanks and take the column numbers above. Fortran writes `E` exponents like `0.12E-04` and `6.1E-024`; most scripting languages read them correctly.

## 7.3 VTK files (ParaView)

`RESULTS_PARA_nn.vtk` is a legacy ASCII `UNSTRUCTURED_GRID` file. Each element and each sub-element of the section/thickness is divided into cells according to the subdivision numbers of the request. The point data are:

**Table 7.3.** Fields in the VTK file (names exactly as written).

| Name          | Type   | Meaning                                  |
|---------------|--------|------------------------------------------|
| Displacements | vector | displacement in the frame of the request |

| Name                                                                                                                                                                                                | Type    | Meaning                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------------------------------|
| <code>Sigma_XX</code> , <code>Sigma_YY</code> , <code>Sigma_ZZ</code> ,<br><code>Sigma_XZ</code> , <code>Sigma_YZ</code> , <code>Sigma_XY</code>                                                    | scalars | stress components                                                                                             |
| <code>VonMises</code>                                                                                                                                                                               | scalar  | equivalent (von Mises) stress                                                                                 |
| <code>Epsilon_XX</code> , <code>Epsilon_YY</code> , <code>Epsilon_ZZ</code> ,<br><code>Epsilon_XZ</code> , <code>Epsilon_YZ</code> , <code>Epsilon_XY</code>                                        | scalars | strain components (engineering shear)                                                                         |
| <code>ID_LAM</code> , <code>ID_MAT</code>                                                                                                                                                           | scalars | lamination id and material id of the sub-element the node belongs to (useful to colour the plies)             |
| <code>Elem.Dimension</code>                                                                                                                                                                         | scalar  | 1, 2 or 3: beam, plate or solid element                                                                       |
| <code>SHEAR</code>                                                                                                                                                                                  | vector  | the coordinates of the node (a quirk of the legacy layout)                                                    |
| <code>ElectricDisplacements</code> ,<br><code>TEMPERATURE_[C]</code> , <code>VOLTAGE[V]</code> , <code>EX</code> , <code>EY</code> ,<br><code>EZ</code> , <code>DOF_SET</code> , <code>HYGRO</code> |         | zeros: placeholders for the fields of the full program, kept so that existing ParaView state files still load |

**In ParaView:** open the file, press *Apply*, choose a field from the colour list (for instance `Sigma_YY`), and apply the filter *Warp By Vector* with `Displacements` (use a *scale factor* so that the deformation is visible) to see the deformed shape. Use the *Cell data* → *point data* default; nodes shared between cells have one value per cell node, so there are visible jumps at element interfaces (the stress field is discontinuous).

#### NOTE

Cells that sample different sub-elements (plies) do not share nodes, so the jump of stress between plies is visible in the plot.

## 7.4 GMSH files

`RESULTS_GMSH_nn.msh` contains the same grid and fields as GMSH 2 views (open it with GMSH; the views appear in the *Modules* → *Post-processing* menu).

## 7.5 Frequencies and modes

`DYNAMIC/FREQUENCIES.dat` :

```
Frequency 1: 8.242627792E+001
Frequency 2: 8.242627792E+001
Frequency 3: 4.963753952E+002
Frequency 4: 4.963753952E+002
```

Frequencies are in **hertz**, ascending. Repeated values mean repeated modes (for example the two bending directions of a square section).

`RESULTS_DYN_PARA.vtk` has one displacement vector field per mode, named like `Mode:001-Freq:0.8243E+02Hz` (mode number and frequency); each mode is normalised with respect to the mass matrix (modal mass 1) and its absolute amplitude is arbitrary: choose a visual scale in ParaView.

`WORK/ARPACK_SOLVER_INFO.dat` lists the Ritz values and the relative residuals of the modes and says which solver was used ( `ARPACK` for large problems, `DENSE` for small ones).

## 7.6 Matrix dumps

If `KMAT` is requested, `WORK/K_MAT.dat` contains one line per stored entry: `row column value`, with the **global DOF numbering of the program: field by field, then node by node, then term** (all `u` DOFs first, then all `v`, then all `w`). For a static analysis the dumped matrix is the matrix **after** the elimination of the constraints (constrained rows and columns are zero with unit diagonal); for the modal analysis it is the full stiffness matrix. `FORCES.dat` and `UNKNOWN.dat` have one value per DOF in the same order.

## 7.7 The warning file

`REPORT/WARNING_file.dat` lists the warnings in order. Typical messages are described in Chapter 10.

## 7.8 Console summary

The lines starting with `[INFO]` summarise the run: the size of the model, the number of Gauss points, the number of non-zeros of  $K$  and the assembly time, the solver time, and the times of input, preprocessing, analysis and output. The line `[ERROR]` (printed on the error stream) explains why a run failed.

# More worked examples

All the examples of this chapter and of the tutorial are generated by

```
python DOC\BOOKS/examples/make_examples.py [output_folder]
powershell DOC\BOOKS\examples\run_examples.ps1 -Cases DOC\BOOKS\examples\cases -Out
runs -Exe MUL2_V3.exe
```

The second command runs every case and prints the tip displacement or the frequencies. The cases share the geometry of the tutorial:  $L = 1$  m,  $0.1 \times 0.1$  m section,  $E = 70$  GPa,  $\nu = 0.3$ ,  $\rho = 2700$  kg/m<sup>3</sup> (the laminate uses the orthotropic material of Chapter 5), clamped at  $Y = 0$ , tip load 1000 N in  $+Z$ .

**Table 8.1.** List of generated cases.

| Case                                       | Model                                              | What it demonstrates                                  |
|--------------------------------------------|----------------------------------------------------|-------------------------------------------------------|
| beam_te{1,2,4}_static/modal                | B4 beam, Taylor 1, 2, 4                            | effect of the expansion order; Poisson locking of TE1 |
| beam_le2_static/modal                      | B4 beam, $2 \times 2$ Q9 Lagrange section          | Lagrange section, no polynomial assumption            |
| plate_strip_static/modal                   | Q9 strip, B3 thickness (Lagrange)                  | a plate model of the same structure                   |
| solid_block_static/modal                   | H27 block                                          | three-dimensional model                               |
| plate_lam_0_90_0_*,<br>plate_lam_90_0_90_* | Q9 strip, three B2 plies with orthotropic material | laminates, ply angles                                 |

## 8.1 Example 1: plate model of the cantilever

The structure is now a *strip of plate*  $0.1 \times 1.0$  m with thickness 0.1 m (thick for a plate, but a fine test). The structural mesh is  $1 \times 8$  Q9 elements in the plane  $Z = 0$ ; every node uses LE; the thickness mesh is one B3 element with three nodes at  $z = -0.05, 0, 0.05$ .

CONNECTIVITY.dat (Q9 plate elements)

```
8
Q9 1  1  2  3  6  9  8  7  4  5  1  1
Q9 2  7  8  9 12 15 14 13 10 11  1  1
Q9 3 13 14 15 18 21 20 19 16 17  1  1
Q9 4 19 20 21 24 27 26 25 22 23  1  1
...  (4 more lines)
```

EXP\_MESH\_01.dat (thickness nodes:  $x = y = 0$ ,  $z$  along the plate normal)

```
3
1  0.0D0 0.0D0 -5.0000000000D-02
2  0.0D0 0.0D0 0.0D0
3  0.0D0 0.0D0 5.0000000000D-02
```

EXP\_CONN\_01.dat (one B3 sub-element)

```
1
B3 1 1  1  2  3
```

VERSORS.dat is `VERSOR 1 1 0 0` (the local  $x$  along the global  $X$ , which is the width of the strip; with the node order used, the plate normal is  $+Z$  and the local  $y$  along the length). The tip load is distributed on the three tip nodes with the Q9 edge weights  $1/6, 4/6, 1/6$ , at  $z = 0$  (the middle node of the thickness mesh).

**Results.** Tip deflection  $5.666 \times 10^{-4}$  m (beam theory with shear:  $5.759 \times 10^{-4}$ ), first frequency 82.48 Hz and the second at 82.50 Hz: the two bending modes are not exactly degenerate (a square section would give equal frequencies) because in a plate model the two bending directions are represented differently: the vertical bending uses the plate plane (surface elements), while the bending in the plane of the strip is carried by the in-plane behaviour of the surface elements.

## 8.2 Example 2: solid model

The structural mesh is  $1 \times 8$  H27 elements; every node uses `LE` with a single-node section and the `s1` expansion element (the expansion mesh is a point). `ANALYSIS.dat` selects `NONE` for solids (never `MITC` with H8; H27 with `MITC` is possible but not necessary here).

**Results.** Tip deflection  $5.663 \times 10^{-4}$  m,  $f_1 = 82.53$  Hz. The solid is the reference "three-dimensional" answer for this problem; the TE4 and LE beam models agree with it within 0.5 % in the deflection and 0.3 % in the frequency, with a fraction of the unknowns for long beams.

### 8.3 Example 3: effect of the order of the Taylor expansion

Run `beam_te1_static`, `beam_te2_static`, `beam_te4_static` and compare. TE1 gives  $4.28 \times 10^{-4}$  m: this is the *Poisson locking* of a linear section kinematics. The error of TE1 is not a bug: it is a property of the model and disappears with TE2. In the beam theories of the textbooks a "Timoshenko beam" is more accurate than TE1 because it has a shear correction factor, which the CUF model does not need when the expansion is rich enough.

### 8.4 Example 4: laminated strip and the angle convention

Compare `plate_lam_0_90_0_static` and `plate_lam_90_0_90_static`: same geometry, same ply properties, two stacking sequences. The tip deflections differ by a factor 8 ( $3.22 \times 10^{-4}$  m versus  $2.67 \times 10^{-3}$  m) and the first frequencies are 120.97 Hz and 49.23 Hz, as discussed in Chapter 5. The stress in each ply can be seen by asking for `PARA` output with a subdivision `1 1 1 1 1 1 1 1 1 1` and colouring by `Sigma_YY` or `ID_LAM`: the three plies appear as three layers of cells; the stress jumps between plies because the plies have different stiffness for the same strain.

To read the normal stress along the thickness at a point, add `PNT` requests at several  $z$  positions slightly inside each ply, for example  $z = -0.04, -0.01, 0.01, 0.04$  at the mid-length section.

### 8.5 Example 5: your own model

A good way to start a new model:

1. Copy the closest example folder.
2. Change the geometry and the connectivity (keep the node order conventions).

3. Run the **modal** analysis first: if the lowest frequencies are reasonable for the structure, units, supports and geometry are probably right.
4. Then run the static analysis with your loads.
5. Refine the mesh and the expansion until the results of interest stabilise.

# Troubleshooting

## 9.1 Reading the messages

The program prints errors on the standard error stream as `[ERROR] ROUTINE: message` and returns exit code 1. Warnings are printed in `REPORT/WARNING_file.dat`. This chapter lists the messages you are likely to meet, with the usual cause and remedy.

## 9.2 Errors

**Table 9.1.** Errors and what to do.

| Message (abridged)                             | Cause                                                                         | Remedy                                                                                            |
|------------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| CANNOT OPEN: INPUT/...                         | a file is missing, or the input directory is wrong                            | check the name of the directory and that all the required files are present (Chapter 4 checklist) |
| INVALID ... count / record / header            | the number of records of a file is not an integer, or a record cannot be read | check the first data line of the file and the number of fields of the failing record              |
| UNKNOWN ELEMENT TOPOLOGY: X                    | the element name is not recognised                                            | use B2 B3 B4 Q4 Q9 Q16 T3 T6 H8 H27                                                               |
| CONNECTIVITY REFERENCES UNKNOWN NODE           | an element uses a node id not in NODES.dat                                    | check the ids                                                                                     |
| DUPLICATE ... ID                               | two records of the same file have the same id                                 | make the ids unique                                                                               |
| SOLUTION n IS NOT SUPPORTED (ONLY 101 AND 103) | ANALYSIS.dat asks for another analysis                                        | use 101 or 103                                                                                    |
| UNKNOWN SHEAR-LOCKING CORRECTION               | a word other than NONE , MITC , REDI , SELI                                   | correct the word                                                                                  |
| THE FIELDS RECORD MUST CONTAIN MECH            | FIELDS without the displacements                                              | add MECH                                                                                          |
| UNKNOWN FIELD IN THE FIELDS RECORD: X          | a word other than MECH , THERMO , PIEZO                                       | correct the word                                                                                  |

| Message (abridged)                                                                                      | Cause                                                                                                                         | Remedy                                                               |
|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| MIXED STRESS FIELDS (RMVT) ARE NOT IMPLEMENTED YET                                                      | STRESS in the FIELDS record                                                                                                   | remove it: the mixed principle is not available                      |
| THERMO IS ACTIVE BUT NO KINEMATICS EXPANDS THE TEMPERATURE (also PIEZO / the potential)                 | the field is switched on in ANALYSIS.dat but KINEMATICS.dat has no expansion for it                                           | give the field an expansion in the kinematics of the nodes           |
| THE JOIN RECORD IS \ JOIN COINCIDENT [TOLERANCE]\` / THE TOLERANCE OF JOIN COINCIDENT MUST BE POSITIVE` | malformed JOIN record                                                                                                         | write JOIN COINCIDENT or JOIN COINCIDENT 1.0E-6                      |
| warning JOIN COINCIDENT FOUND NO COINCIDENT DEGREES OF FREEDOM                                          | no DOF of different nodes is at the same point (within the tolerance)                                                         | check the coordinates and the section meshes; increase the tolerance |
| ONLY LAM2 IS ACTIVE FOR SOLUTIONS 101/103                                                               | another lamination type in LAMINATION.dat                                                                                     | use LAM2 id material angle_y angle_z                                 |
| UNSUPPORTED NODE MODEL (ONLY TE, LE AND HLE)                                                            | another token in the historical NODES.dat                                                                                     | use TE n, LE or HLE                                                  |
| MISCELLANEOUS BASIS IS NOT IMPLEMENTED YET                                                              | Mn in KINEMATICS.dat                                                                                                          | use TE n, LE or HLE                                                  |
| HLE NODES NEED A HLE SECTION MESH (HQ4/HB2) AND LE NODES A LAGRANGE ONE                                 | HLE node on a Lagrange mesh or LE node on an HQ4/HB2 mesh                                                                     | match nodes and mesh; join the two with a Taylor node                |
| ONLY U, V AND W ARE SUPPORTED BY 101/103                                                                | a field other than U, V, W is active in a kinematic                                                                           | set the other fields to NONE                                         |
| LE NODE USES INCOMPATIBLE EXPANSIONS                                                                    | the elements that share a Lagrange node use different expansion meshes                                                        | give them the same section number, or use Taylor                     |
| NODE HAS INCOMPATIBLE EXPANSION DIMENSIONS, EXPANSION HAS MIXED DIMENSIONS                              | a node is shared by elements whose expansion meshes have different dimensions, or the sub-elements of one mesh mix dimensions | do not mix families at a node; check EXP_CONN                        |
| ELEMENT REFERENCES UNKNOWN EXPANSION / SOR                                                              | the section number or versor id of an element does not exist                                                                  | add the files/records                                                |
| SOR IS PARALLEL TO BEAM AXIS                                                                            | the versor is parallel to the axis of a beam                                                                                  | choose another versor (Chapter 5)                                    |

| Message (abridged)                                                                         | Cause                                                                                                                                                      | Remedy                                                                                                                    |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| DEGENERATE BEAM AXIS ,<br>DEGENERATE SURFACE , SOR IS<br>NORMAL TO SURFACE                 | first and last node coincide; three<br>corners are collinear; the versor is<br>normal to a plate                                                           | correct the coordinates or<br>the versor                                                                                  |
| EXPANSION MESH IS<br>DEGENERATE                                                            | the nodes of an expansion mesh<br>do not span the section (beam) or<br>thickness (plate)                                                                   | check EXP_MESH<br>coordinates                                                                                             |
| NO CONSTRAINED DEGREE OF<br>FREEDOM: THE SYSTEM IS<br>SINGULAR                             | no D-PLANE selected a DOF in a<br>static analysis                                                                                                          | add a support                                                                                                             |
| PARTIAL PLANE CONSTRAINT<br>REQUIRES MPC FOR TE                                            | a plane cuts the section of a Taylor<br>node                                                                                                               | constrain a plane that<br>contains the whole<br>section, or use LE                                                        |
| CONFLICTING PRESCRIBED<br>VALUES                                                           | two constraints prescribe different<br>values for the same DOF                                                                                             | remove the conflict                                                                                                       |
| POINT LOAD LOCATION WAS<br>NOT FOUND                                                       | the load is not at a section node of<br>a structural node                                                                                                  | use the exact coordinates<br>of a section node<br>(structural node + section<br>node coordinates in the<br>element frame) |
| LOAD ACTS ON AN INACTIVE<br>FIELD                                                          | a load component on a field that<br>is NONE                                                                                                                | check the kinematic                                                                                                       |
| PARDISO FACTORIZATION<br>ERROR CODE -4 (ZERO PIVOT:<br>SINGULAR MATRIX, CHECK<br>SUPPORTS) | the matrix is singular: not enough<br>supports, a part of the model not<br>connected or not supported,<br>MITC on H8, or a nearly singular<br>Taylor basis | check the supports and<br>the connectivity; remove<br>MITC from H8; reduce<br>the Taylor order or rescale<br>the section  |
| PARDISO ... ERROR CODE -2<br>(NOT ENOUGH MEMORY)                                           | the factor does not fit in memory                                                                                                                          | reduce the model, use a<br>lower expansion order                                                                          |
| K-SIGMA*M CANNOT BE<br>FACTORED ... (MISSING<br>SUPPORTS?)                                 | modal analysis of a structure with<br>rigid-body modes                                                                                                     | add supports                                                                                                              |
| THE SOLUTION CONTAINS NAN<br>OR INFINITY                                                   | numerical failure after the solution                                                                                                                       | check units, supports and<br>Taylor order                                                                                 |

### 9.3 Warnings

**Table 9.2.** Warnings.

| Message (abridged) | Meaning |
|--------------------|---------|
|--------------------|---------|

| Message (abridged)                                                               | Meaning                                                                                                                                         |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| EXTRA RECORDS AFTER DECLARED COUNT                                               | a file has more records than the first line says; the extras are ignored (check the count)                                                      |
| FEWER RECORDS THAN DECLARED: THE REST IS IGNORED                                 | POSTPROCESSING.dat declares more records than present                                                                                           |
| THE SOLVER RECORD OF ANALYSIS.DAT IS OBSOLETE: IGNORED                           | the file has the old solver line; delete it (harmless)                                                                                          |
| REDI/SELI IS NOT DEFINED FOR TRIANGLES: FULL INTEGRATION                         | T3/T6 elements have no reduced rule; they were integrated fully                                                                                 |
| REDI LEAVES HOURGLASS MODES IN Q4, H8 AND H27: USE SELI                          | zero-energy modes can make the matrix singular (PARDISO error) or the solution meaningless: use SELI or MITC                                    |
| MITC IS NOT DEFINED FOR SOME TOPOLOGY: FULL INTEGRATION                          | the element has no tying table; full integration was used (risk of locking)                                                                     |
| NON-MECHANICAL OR NONSYMMETRIC DATA WERE PRESERVED                               | the material file has thermal/electrical records or a non-symmetric MAT-C                                                                       |
| UNUSED NODES HAVE NO DEGREES OF FREEDOM                                          | some nodes of NODES.dat belong to no element                                                                                                    |
| PLANE DID NOT SELECT ANY ACTIVE DOF                                              | a D-PLANE plane does not touch any node or section node: probably a sign or coordinate mistake (remember $AX + BY + CZ + D = 0$ )               |
| LOADS AND NON-ZERO PRESCRIBED DISPLACEMENTS ARE IGNORED                          | analysis 103 ignores loads and imposed displacements; normal                                                                                    |
| MORE MODES REQUESTED THAN FREE DEGREES OF FREEDOM                                | the number of modes was reduced                                                                                                                 |
| NEGATIVE EIGENVALUES: K IS NOT POSITIVE SEMI-DEFINITE                            | rounding or rigid-body modes; check the supports                                                                                                |
| CHOLSKY FAILED (ILL-CONDITIONED MATRIX): USED SYMMETRIC INDEFINITE FACTORIZATION | the matrix is very ill-conditioned (typically high-order Taylor); the results are probably still good, but <b>verify</b> with another expansion |
| POINT n IS OUTSIDE THE MODEL: NO RESULT                                          | a PNT point is not inside any element; the row of POST_POINT.dat contains NaN                                                                   |

## 9.4 Symptoms without a message

**Table 9.3.** Strange results.

| Symptom | Likely cause |
|---------|--------------|
|---------|--------------|

| Symptom                                                        | Likely cause                                                                                                                         |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| displacements of the wrong sign or direction                   | the <b>element frame</b> is not what you think: versor, node order (plate normal) — Chapter 5 recipes                                |
| the structure seems far too stiff                              | TE 1 ; TE 0 ; missing section terms; locking ( NONE on a thin plate; B2/B3 without MITC ); a constraint that is too severe           |
| the structure seems far too flexible                           | missing or wrong supports (a plane with the wrong sign of <i>D</i> ); a wrong unit (E in GPa instead of Pa)                          |
| laminates stiffness does not change with the stacking sequence | angles are in degrees, applied to the <b>lamination</b> that each sub-element refers to; check EXP_CONN lamination ids               |
| strange wiggles in the VTK display                             | PARA output with 8 nodes per cell on a high-order model; use 20/27 nodes and subdivisions                                            |
| point result is NaN                                            | the point is outside the structure; check the global coordinates                                                                     |
| high-order Taylor results change when the order grows          | conditioning: rescale the section or use LE                                                                                          |
| modal frequencies too low by orders of magnitude               | units of density or modulus; check consistency                                                                                       |
| the run takes too long or uses too much memory                 | the number of DOF is larger than expected; use the console line PREPROCESSING: ... DOF to check, and Chapter 6 for the cost estimate |

## 9.5 Where to look for more information

- The console output and REPORT/WARNING\_file.dat (always).
- WORK/ARPACK\_SOLVER\_INFO.dat for the convergence of the modes.
- Run with MUL2\_DEBUG=1 for extra lines.
- The Theoretical Guide explains the assumptions of every model; the Implementation Guide explains where each message is generated.

# Hierarchical sections (HLE)

HLE refines a section by the **order**  $p$  of its sub-elements instead of the number of nodes. A few sub-elements of order 3–5 usually describe a section better than a fine Lagrange mesh, and curved walls (tubes) are described exactly. Theory: Chapter 14 of the Theoretical Guide.

## 10.1 Input

**NODES.dat.** The structural nodes use the token `HLE` (no order):

```
1  0.0D0  0.0D0  0.0D0  HLE
```

or `HLE` in `KINEMATICS.dat`.

**EXP\_MESH\_nn.dat.** Only the **vertices** of the sub-elements are listed (and, for curved sides, the extra mid nodes).

**EXP\_CONN\_nn.dat.**

```
HQ4 id lamination v1 v2 v3 v4 p [m1 m2 m3 m4]
HB2 id lamination v1 v2 p
```

`HQ4` is a quadrilateral (vertices counter-clockwise, like Q4), `p` its order (1–8 tested); `HB2` is a line for the thickness of a plate. The optional `m1...m4` are mid nodes of the sides 1–2, 2–3, 3–4, 4–1: a mid node off the chord makes that side a circular arc, a mid node on the chord a straight side.

Orders may differ from one sub-element to the next. On a shared side the modes up to the smaller order are shared; nothing else has to be specified, and the numbering direction of the sub-elements is free.

## 10.2 Example: a tube

Four quarter-annulus sub-elements of order 4 (outer radius 0.5, inner 0.4):

EXP\_MESH\_01.dat (20 nodes)

```
1  0.4  0  0
2  0.5  0  0
3  0.0  0  0.4
...
```

EXP\_CONN\_01.dat (4 elements)

```
HQ4 1 1 1 2 4 3 4 100 200 101 300
HQ4 2 1 3 4 6 5 4 101 201 102 301
HQ4 3 1 5 6 8 7 4 102 202 103 302
HQ4 4 1 7 8 2 1 4 103 203 100 303
```

(ids 1–8 are the vertices at 0°, 90°, 180°, 270°, inner then outer on each ray; 100–103 the mid-radius points on the rays, 200–203 the midpoints of the outer arcs and 300–303 of the inner arcs. Coordinates are local: columns  $x$ ,  $y = 0$ ,  $z$ .) The tube gives a first bending frequency within 1% of Timoshenko beam theory with four elements.

## 10.3 Constraints and loads

- A **D-PLANE** that contains the whole section constrains all the modes. A plane that cuts the section constrains the vertices on it and the side modes whose two end vertices lie on it; with curved sides cut by the plane this is only approximate.
- A point load must be at a vertex; it is applied to the vertex mode. Side and internal modes receive no load, so for  $p \geq 2$  a point load is not the exact equivalent of a traction: compare stresses away from the loaded section.

## 10.4 Mixing with other expansions

An HLE node needs sub-elements of kind HQ/HB, a Lagrange node a Lagrange mesh. Taylor nodes work with any mesh, so a beam can go from HLE to LE (or TE) through a **Taylor node** on the interface, and TE and HLE nodes may share an element.

## 10.5 Limits

No triangles and no solids; circular arcs only; HQ4 and HB2 only. The static viewer of the interface draws a curved side as a polyline through its mid node.

The interface (**EXP\_CONN** editor) has an HLE column and the buttons *+ Add HQ4* and *+ Add HB2*.

# Multiphysics: piezoelectric and thermal problems

MUL2\_NEW solves, with the same elements, expansions and analyses, the fields that the historical code called  $T$  (temperature) and  $P$  (electric potential), together with the displacements. No new analysis type is needed: the physics that are solved are chosen in `ANALYSIS.dat` (record `FIELDS`, below); `KINEMATICS.dat` only defines **how each field is expanded**.

## 11.1 Choosing the physics

## 11.2 Choosing the physics: the `FIELDS` record

The last, optional record of `ANALYSIS.dat` lists the physics that are solved:

`FIELDS MECH PIEZO`

| Token  | Field                                                                                 |
|--------|---------------------------------------------------------------------------------------|
| MECH   | displacements $u, v, w$ (always required)                                             |
| THERMO | temperature $T$                                                                       |
| PIEZO  | electric potential $P$                                                                |
| STRESS | transverse stresses of the mixed principle (RMVT): <b>reserved</b> , an error for now |

A field that is not listed is not solved, even if `KINEMATICS.dat` gives it an expansion: the same input switches the physics on and off by changing one line. A field that is listed must be expanded in at least one kinematics (otherwise the run stops with an explicit message). **Without the record** every field expanded in `KINEMATICS.dat` is solved, as before. A boundary condition on a field that is switched off (for instance a voltage with `FIELDS MECH`) has no effect.

The principle of the element (displacement-based, PLV, today) is not an analysis option: the mixed Reissner principle (RMVT) will be selected element by element with a label in `CONNECTIVITY.dat`, and an RMVT element will bring its own stress

KINEMATICS.dat has nine expansion tokens per record, in the order  $u, v, w, T, P, B, \sigma_{zz}, \sigma_{xz}, \sigma_{yz}$ . A field with NONE is not solved.

1

|             |    |    |    |      |      |      |      |      |      |                 |
|-------------|----|----|----|------|------|------|------|------|------|-----------------|
| KINEMATIC 1 | LE | LE | LE | NONE | LE   | NONE | NONE | NONE | NONE | ! piezoelectric |
| KINEMATIC 2 | LE | LE | LE | LE   | NONE | NONE | NONE | NONE | NONE | ! thermoelastic |
| KINEMATIC 3 | LE | LE | LE | LE   | LE   | NONE | NONE | NONE | NONE | ! thermo-piezo  |

Temperature and potential accept the Lagrange (LE), hierarchical (HLE) and Taylor (TE) expansions like the displacements. The nodes of a model may mix them. The temperature is a **change** with respect to the stress-free reference state.

| Active fields | Problem           | 101              | 103         | 104             | 105              | 106 | 108 |
|---------------|-------------------|------------------|-------------|-----------------|------------------|-----|-----|
| $u, v, w$     | structural        | yes              | yes         | yes             | yes              | yes | yes |
| $+P$          | piezoelectric     | yes              | yes (SC/OC) | yes             | yes (condensed)  | yes | yes |
| $+T$          | thermoelastic     | yes (monolithic) | no          | yes (transient) | yes (pre-stress) | yes | yes |
| $+T + P$      | pyro/thermo-piezo | yes              | no          | yes             | yes              | yes | yes |

## 11.3 Materials

All the records are in `MATERIAL.dat`; the header is `number_of_materials` `number_of_records` and every record refers to a material by its identifier. The records of the same material may appear in any order.

| Record      | Values | Meaning                                                                                                                                                              |
|-------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Z-EXP<br>id | 18     | piezoelectric stress coefficients $e$ [C/m <sup>2</sup> ], 3 rows (electric direction $T, L, Z$ ) by 6 columns (Voigt $xx, yy, zz, xz, yz, xy$ of the material axes) |
| Z-PRM<br>id | 9      | permittivity [F/m] at constant strain, row by row                                                                                                                    |
| T-EXP<br>id | 6 or 1 | thermal expansion [1/K]: $\alpha_{TT}, \alpha_{LL}, \alpha_{ZZ}, \alpha_{TZ}, \alpha_{LZ}, \alpha_{TL}$ , or a single isotropic value                                |
| T-CON<br>id | 9 or 1 | heat conductivity [W/(m K)], row by row, or a single isotropic value                                                                                                 |

| Record      | Values | Meaning                                                                         |
|-------------|--------|---------------------------------------------------------------------------------|
| T-SPC<br>id | 1      | specific heat [J/(kg K)]; the heat capacity is $\rho c$ (needed by 104 and 106) |
| PIROE<br>id | 3      | pyroelectric coefficient [C/(m <sup>2</sup> K)] <b>at zero stress</b>           |
| DAMP        | 2      | Rayleigh damping, DAMP k_K k_M: $C = k_M M + k_K K$ (no identifier)             |

Example of a piezoelectric ceramic polarised along Z (PZT-5H like):

|         |                                                                    |
|---------|--------------------------------------------------------------------|
| 1 3     |                                                                    |
| MAT-C 1 | 127.2D9 80.2D9 84.7D9 0 0 0 80.2D9 127.2D9 84.7D9 0 0 0 ... 7500.0 |
| Z-EXP 1 | 0 0 0 0 17.03 0 0 0 0 17.03 0 0 -6.62 -6.62 23.24 0 0 0            |
| Z-PRM 1 | 1.503D-8 0 0 0 1.503D-8 0 0 0 1.300D-8                             |

The material axes follow the lamination angles like the stiffness: the coefficients are rotated into the element frame (tensors  $e$ ,  $\epsilon$ ,  $\alpha$ ,  $k$  and the vector  $p$ ).

**Pyroelectricity.** PIROE is the coefficient measured on a free body ( $D = pT$  with  $E = 0$  and no stress). The code subtracts the part that comes from the thermal expansion,  $p_\epsilon = p_\sigma - e \alpha$ , so that the input is the quantity found in the data sheets.

## 11.4 Boundary conditions

All the planes are  $ax + by + cz + d = 0$ , as in D-PLANE. A record with a plane that contains the whole section (or the whole model) acts on all the expansion nodes of the plane.

| Record                           | Effect                                                                                                                                               |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| V-PLANE id a b c<br>d V          | electric potential $V$ [V] on a plane (electrode)                                                                                                    |
| V-POINT id x y z<br>V            | potential at a section node                                                                                                                          |
| V-FLOAT id a b c<br>d            | floating electrode: all the potential degrees of freedom of the plane are tied to one unknown value (open circuit, sensor). Supported by 101 and 103 |
| F-POINT id x y z<br>fx fy fz [Q] | the optional last value is an electric charge                                                                                                        |
| T-PLANE id a b c<br>d T          | temperature on a plane                                                                                                                               |

| Record                      | Effect                                                                                                                                      |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| T-POINT id x y z<br>T       | temperature at a section node                                                                                                               |
| T-CONST id T                | uniform temperature everywhere (thermal load without conduction)                                                                            |
| Q-POINT id x y z<br>P       | heat power [W] at a section node                                                                                                            |
| Q-PLANE id a b c<br>d q     | heat flux $q$ [W/m ^ 2] <i>into</i> the exposed surface of a plane                                                                          |
| Q-SUN id sx sy<br>sz G a    | solar load on the whole exposed skin: $q = a G \max(0, n \cdot s)$ ; $s$ points towards the sun, $G$ [W/m ^ 2] irradiance, $a$ absorptivity |
| Q-CONV id a b c<br>d h Tinf | convection $q = h(T_{\infty} - T)$ on the exposed surface of a plane; the plane $\emptyset \emptyset \emptyset$ means all the skin          |

Every value record ( D-PLANE , V-PLANE , V-POINT , T-PLANE , T-POINT , T-CONST , Q-PLANE ) may end with the number of a **field** of `FIELDS.dat` (next section): the value is multiplied by the field at the position of every node (or integration point).

**The exposed skin.** Surface loads act on the surfaces that are not shared with another element: the lateral surface and the ends of a beam, top, bottom and sides of a plate, the faces of a solid. The outward normal points away from the element. The sun does not cast shadows: every exposed face with  $n \cdot s > 0$  is lit. There is no radiation to the environment (use `Q-CONV`).

With only `Q-SUN` and `Q-CONV` a steady solution exists without any prescribed temperature: the convection makes the conduction matrix definite. The console log reports for every surface load the area it selected and the power it applies.

## 11.5 Fields (FIELDS.dat)

A field is a function  $F(x, y, z)$  written as a sum of terms. The optional file has the number of fields, then, for every field, a header and one line per term.

```

2

FIELD 1 1
Y-EXP 1.0D0 1.0D0           ! F = y

FIELD 2 2
CONST 1.0D0
SIN-X 0.5D0 6.2832D0 0.0D0   ! F = 1 + 0.5 sin(2 pi x)

```

| Term                | Constants         | Value                                                   |
|---------------------|-------------------|---------------------------------------------------------|
| CONST               | $c_1$             | $c_1$                                                   |
| X-EXP, Y-EXP, Z-EXP | $c_1 \ c_2$       | $c_1 x^{c_2}$ (idem $y, z$ )                            |
| COS-X, COS-Y, COS-Z | $c_1 \ c_2 \ c_3$ | $c_1 \cos(c_2 x + c_3)$                                 |
| SIN-X, SIN-Y, SIN-Z | $c_1 \ c_2 \ c_3$ | $c_1 \sin(c_2 x + c_3)$                                 |
| ATNZX               | $c_1 \ c_2 \ c_3$ | $c_1 \operatorname{atan2}(z - c_2, x - c_3)$ in degrees |
| B-SIN               | $c_1 \dots c_5$   | $c_1 \sin(c_2 \pi x / c_3) \sin(c_4 \pi y / c_5)$       |

A traveling-wave or sinusoidal voltage distribution on an electrode, a temperature gradient or a spatially varying heat flux are examples. A reference to a missing field is read as 1.

## 11.6 Results

- `STATIC/POST_POINT.dat`: the column `DT` is the temperature, `VOLT` the potential, `E11...D33` the electric field and displacement, and three more columns at the end give the heat flux  $q = -k \nabla T$  (global frame).
- The VTK files hold `TEMPERATURE[C]`, `VOLTAGE[V]`, the electric displacement, `EX/EY/EZ` and the vector `HeatFlux`.
- The stresses are the mechanical ones,  $\sigma = C(\epsilon - \alpha T) + \dots$ : the thermal strain is not part of the stress.

## 11.7 Checks you can repeat

| Test                                      | Closed form                                                                    | Error       |
|-------------------------------------------|--------------------------------------------------------------------------------|-------------|
| free piezoelectric slab, 100 V            | $u_z = d_{33} V, E = -V / t$                                                   | $< 10^{-6}$ |
| thickness resonance, short / open circuit | piezoelectric stiffening $c + e^2 / \epsilon$                                  | 0.03 % (OC) |
| bar with a uniform temperature rise       | $u = \alpha \Delta T L$ , no stress; clamped:<br>$\sigma = -E \alpha \Delta T$ | $10^{-9}$   |
| conduction between two temperatures       | linear profile, $q = -k \Delta T / L$                                          | exact       |
| heat power at the end of a bar            | $T = QL / (kA)$                                                                | $10^{-9}$   |
| convection in series with conduction      | $T_L = hT_\infty / (k / L + h)$                                                | $10^{-9}$   |

| Test              | Closed form                                                      | Error             |
|-------------------|------------------------------------------------------------------|-------------------|
| pyroelectric slab | closed circuit $D = pT$ ; open circuit<br>$V = pTt / \epsilon^T$ | $5 \cdot 10^{-7}$ |

The scripts are in TESTS/PIEZO , TESTS/THERMAL and TESTS/DYNAMICS .

# 11.8 Limits

The floating electrode (V-FLOAT ) works in 101 and 103 only. The temperature is not solved by 103 (modal analysis: use 106 for the dynamic behaviour of a thermo-mechanical model). Q-SUN has no shadowing and no emission. FIELDS.dat is not used for laminate angles, point forces or F-PRESS .

# Time, buckling, frequency and nonlinear analyses

ANALYSIS.dat selects the analysis with its first record. The other records (number of modes, shear-locking treatments) keep their meaning.

| Id  | Analysis                       | Extra input file | Main output                                                        |
|-----|--------------------------------|------------------|--------------------------------------------------------------------|
| 101 | linear static                  |                  | STATIC/                                                            |
| 103 | free vibration                 |                  | DYNAMIC/FREQUENCIES.dat , mode shapes                              |
| 104 | time response (Newmark)        | TIME_RESP.dat    | DYNAMIC/TIME_HISTORY.dat ,<br>DYNAMIC/RESULTS_PARA_nn_Ssssssss.vtk |
| 105 | linear buckling                |                  | DYNAMIC/BUCKLING_FACTORS.dat , modes                               |
| 106 | frequency response             | FREQ_RESP.dat    | DYNAMIC/FREQ_HISTORY.dat                                           |
| 108 | geometrically nonlinear static | NL_INFO.dat      | DYNAMIC/TIME_HISTORY.dat , STATIC/                                 |

The loads are the ones of BC.dat (forces, charges, heat, surface loads) and the prescribed values (displacements, potentials, temperatures) in all the analyses; the extra file says *how they vary*.

## 12.1 Time response (104)

### 12.1.1TIME\_RESP.dat

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| 0.0 0.5             | TI TF                                               |
| 1000                | NSTEP                                               |
| 10                  | output every 10 steps                               |
| 0                   | number of extra output steps (their indices follow) |
| -----               |                                                     |
| 1                   | number of time loads                                |
| STEP 0.0 1.0        |                                                     |
| NEWMARK 0.25 0.5    | (optional)                                          |
| RAYLEIGH 0.0 1.0D-4 | (optional) alpha beta C = alpha M + beta K          |
| THETA 1.0           | (optional) thermal theta-method, 0.5 .. 1           |
| T0 293.0            | (optional) reference temperature [K]                |

Dashed separator lines are ignored. The **amplitude**  $a(t)$  is the sum of the load records and multiplies *every* load and prescribed value of `BC.dat` (a voltage, a temperature, a force). The initial state is  $a(0)$  times the prescribed values, with zero velocity and zero temperature.

| Record                             | $a(t)$                                                       |
|------------------------------------|--------------------------------------------------------------|
| STEP $t_0$ A                       | $A$ for $t \geq t_0$                                         |
| RAMP $t_0$ $t_1$ A                 | linear from 0 to $A$ between $t_0$ and $t_1$ , then $A$      |
| IMPU $t_0$ A                       | $A$ at the first step $t \geq t_0$ only                      |
| SINU $\phi$ $\omega$ A             | $A \sin(\phi + \omega t)$                                    |
| HASU $t_0$ F $\phi_2$ n $\phi_1$ A | sine burst of frequency $F$ with a Hann window of $n$ cycles |
| HACU $t_0$ F $\phi_2$ n $\phi_1$ A | the same with a cosine                                       |
| WPSU $t_0$ F $\phi_2$ n $\phi_1$ A | sine burst with a sin2 window                                |

**Damping.** The Rayleigh damping acts on the mechanical degrees of freedom. It may be given by `RAYLEIGH alpha beta` here or by the historical record `DAMP k_K k_M` of `MATERIAL.dat` (note the order:  $K$  first). For a mode of frequency  $\omega$  the damping ratio is  $\zeta = \frac{1}{2} \alpha / \omega + \beta \omega$ .

**Integration.** Mechanical degrees of freedom: Newmark (default  $\beta = 1/4$ ,  $\gamma = 1/2$ , average acceleration, unconditionally stable and without numerical damping). Temperature: first-order  $\theta$ -method (default backward Euler;  $\theta = 1$ ;  $\theta = 1/2$  is Crank-Nicolson and needs a smooth start). Potential: no inertia (algebraic). The effective matrix is factored **once**.

**Thermal coupling.** With `T-SPC` the temperature has a heat capacity  $\rho c$  and the heat equation is transient. The temperature acts on the structure through the thermal expansion. The reverse coupling (thermoelastic heating,  $T_0 \beta: \epsilon$ , the cause of thermoelastic damping) is added when `T0` is given.

### 12.1.2 Output

`TIME_HISTORY.dat` has one row per output step and point of `POSTPROCESSING.dat`: the time followed by the columns of `POST_POINT.dat`. Every `PARA` record writes a VTK file per output step.

## 12.2 Buckling (105)

The reference loads (and thermal loads) of the model are applied, the linear static problem is solved and the geometric stiffness  $K_G$  of its stress field is built; the eigenproblem

$$(K + \lambda K_G) x = 0$$

gives the **load factors**  $\lambda$ : the critical load is  $\lambda$  times the applied one. The number of factors is the *number of modes* of ANALYSIS.dat. BUCKLING\_FACTORS.dat lists them in ascending order and the mode shapes are written like the vibration modes. The reference load must be compressive to give a positive factor.

The eigensolver is dense: at most 8000 free degrees of freedom. The temperature is the given field of the pre-stress (a **thermal buckling** problem is a model with a thermal load), the potential is condensed (closed circuit with the prescribed electrodes). Floating electrodes are not supported.

## 12.3 Frequency response (106)

|                   |                 |
|-------------------|-----------------|
| 0.0 400.0         | FI FF [Hz]      |
| 400               | number of steps |
| 1                 | output every n  |
| RAYLEIGH 12.0 0.0 | (optional)      |
| T0 293.0          | (optional)      |

The system  $(K - \omega^2 M + i\omega C) x = F$  is solved on  $N + 1$  frequencies, with the loads of BC.dat as unit harmonic amplitudes (phase 0) and the prescribed values at phase 0. The temperature rows get  $i\omega\rho c$  and the thermoelastic heating (with T0). FREQ\_HISTORY.dat has for every frequency and point the real and imaginary parts of  $u_x, u_y, u_z, T$  and the potential: the modulus is  $\sqrt{\text{Re}^2 + \text{Im}^2}$  and the phase  $\text{atan2}(\text{Im}, \text{Re})$ . Each frequency is a new factorization of a system twice as large as the static one.

## 12.4 Geometrically nonlinear static (108)

|        |                                              |
|--------|----------------------------------------------|
| 1      | technique (1 = load control, 2 = arc length) |
| 10     | number of load steps                         |
| 30     | maximum iterations per step                  |
| 1.0D-8 | relative residual tolerance                  |
| 1      | output every n steps                         |

The loads and prescribed values are applied in  $N$  equal increments. The formulation is total Lagrangian with the Green strain and the second Piola-Kirchhoff stress of the (St. Venant-Kirchhoff) material; the electric potential and the temperature are solved in the same system, so thermal strains, the pyroelectric effect and the piezoelectric coupling are present. Loads are **dead** (they keep their direction). The tangent matrix is rebuilt and factored at every iteration. `TIME_HISTORY.dat` holds the load factor  $\lambda$  in its time column; the standard `STATIC/` files hold the final state (strain and stress there are the linearised measures of the final displacements).

A prescribed displacement is a displacement-controlled step. With technique 1 a limit point (snap-through under load control) cannot be followed; a loss of convergence is reported with the step and the residual.

### 12.4.1 Arc length (technique 2)

With technique 2 the load factor  $\lambda$  becomes an unknown and the equilibrium path is followed through limit points (snap-through, snap-back). The method is Crisfield's with a cylindrical constraint on the displacement degrees of freedom,  $\|\Delta u\| = \Delta s$ ; the reference load ( $\lambda = 1$ ) is the force of the model. The record `number of load steps` is now the **maximum number of arc-length steps**, and a sixth record is added:

|           |                                       |
|-----------|---------------------------------------|
| 2         | technique: arc length                 |
| 100       | maximum number of steps               |
| 30        | maximum iterations per step           |
| 1.0D-8    | relative residual tolerance           |
| 1         | output every n steps                  |
| 0 0 0 3.0 | ds0 dsmin dsmax lambda_max (optional) |

`ds0`, `dsmin`, `dsmax` equal to 0 mean automatic values (the first step is 10 % of  $\lambda_{max}$  in the linear regime,  $\Delta s_{min} = 10^{-4} \Delta s_0$ ,  $\Delta s_{max} = 10 \Delta s_0$ ). After each step  $\Delta s$  is multiplied by  $\sqrt{5 / n_{it}}$  (limited by `dsmin`, `dsmax`); a step that does not converge is repeated with half the arc length. The path ends when  $\lambda \geq \lambda_{max}$  (default 1) or after the maximum number of steps;  $\lambda$  may become negative on the way (snap-back, inverted snap-through). `TIME_HISTORY.dat` has  $\lambda$  in the time column, one row per output step.

Restrictions: all the prescribed values (displacements, potentials, temperatures) must be zero, because the load is the parameter; the norm of  $\Delta u$  uses only the displacement unknowns. Chosen  $\lambda_{max}$  below the first limit load, the arc-length path coincides with load control (checked to  $10^{-9}$ ).

## 12.5 Checks you can repeat

| Test                               | Reference                                             | Error                              |
|------------------------------------|-------------------------------------------------------|------------------------------------|
| step load on a cantilever          | peak $2u_s$ , period $= 1 / f_1$                      | $2.4 \%$ , $10^{-4}$               |
| Rayleigh damping                   | overshoot $e^{-\pi\zeta / \sqrt{1-\zeta^2}}$          | $5 \%$                             |
| heat conduction with capacity      | series solution                                       | $< 0.3 \%$                         |
| lumped bar with sun and convection | $T = \frac{aGA}{hS}(1 - e^{-t/\tau})$                 | see <code>dynamics_tests.py</code> |
| thermoelastic damping              | decay only with $T_0$                                 | $6.5 \%$ in 2 periods              |
| frequency response                 | resonance at $f_1$ , amplitude $\approx 1 / (2\zeta)$ | $< 1 \%$                           |
| buckling of a cantilever           | $\pi^2 EI / 4L^2$                                     | $0.06 \%$                          |
| clamped-clamped bar, thermal       | $4\pi^2 EI / L^2$ with the shear term                 | $5 \%$                             |
| elastica $PL^2 / EI = 1, 2$        | Bisshopp-Drucker                                      | $0.3 \%$ , $0.06 \%$               |
| beam-column, $P = P_{cr} / 2$      | $3(\tan u - u) / u^3$                                 | $0.7 \%$                           |
| thermal expansion, Green measure   | $\sqrt{1 + 2\alpha\Delta T} - 1$                      | $10^{-10}$                         |

The scripts are in `TESTS/DYNAMICS` , `TESTS/BUCKLING` and `TESTS/NONLINEAR` .

## 12.6 Limits

- Floating electrodes (`V-FLOAT`) work only in 101 and 103.
- 104 has a constant time step, zero initial conditions and one amplitude function for all the loads.
- 105 is dense (8000 free degrees of freedom) and the buckling problem is the one of a perfect structure: no post-buckling.
- 108 has an arc-length method (technique 2) only for loads with zero prescribed values; it has no material nonlinearity and no follower loads; MITC ties only the linear part of the strain.

# Curved beams and shells

The beam elements B2 B3 B4 and the plate elements Q4 Q9 Q16 of the previous chapters have **one straight axis** or **one flat plane** per element. This chapter describes the same families when the geometry is *curved*: a curved beam (arch, ring, frame of a fuselage) and a **shell** (skin of a wing, of a fuselage, a tube, a dome). There is no new element name to learn and **no new input file** is required for the usual case; the nodes decide.

## 13.1 What you write

| You write                                               | The program does                                                |
|---------------------------------------------------------|-----------------------------------------------------------------|
| B3, B4 with the nodes <b>on a curve</b> (not on a line) | curved beam: the axis is the polynomial curve through the nodes |
| Q4, Q9, Q16 with the nodes <b>out of a plane</b>        | shell: the surface is the polynomial surface through the nodes  |
| CB2 CB3 CB4, S4 S9 S16 instead of B2 B3 B4, Q4 Q9 Q16   | the same topology, with the curved geometry <b>forced</b>       |

A node deviation larger than  $10^{-6}$  of the length (beam) or of the diagonal (plate) from the line or from the plane makes an element curved. Flat plates and straight beams keep the previous formulation (and the previous results to the last digit). Use S9 for the flat panels of a model that also has curved ones: all the shells then share the same formulation (MITC9).

The *section* of a beam and the *thickness* of a shell are described, as before, by the expansion mesh of the element ( EXP\_MESH\_nn.dat , EXP\_CONN\_nn.dat ) and by the kinematics of the nodes ( TE n , LE , HLE , KINEMATICS.dat ):

| Element     | Expansion mesh                                     | Where it lives                            |
|-------------|----------------------------------------------------|-------------------------------------------|
| curved beam | 2D (the section)                                   | in the plane normal to the axis           |
| shell       | <b>1D (the thickness)</b> , exactly as for a plate | along the director, normal to the surface |

**The thickness is therefore the extent of the 1D expansion mesh.** It is not a separate input and it does not have to be constant in the model: each zone uses its own expansion mesh ( `expansion_id` of the element). The geometry of an element is the mid-surface of the nodes plus the position in the expansion mesh along the director; so a thickness step between two zones is a real step of the faces; the mid-surface is continuous. Any expansion ( `TE` of any order, `LE` layer-wise with one or more layers, `HLE` ) can be used for the thickness.

## 13.2 Reference vector ( `VERSORS.dat` )

The record `frame_id` of the element selects a vector in `VERSORS.dat`, as for the straight elements. For curved elements it is used at **every point**:

- curved beam: the vector is projected on the plane normal to the axis and gives the local axis  $e_3$  of the section (as for a straight beam). It must not be parallel to the axis anywhere on the element;
- shell: the vector is projected on the tangent plane and gives the local axis  $e_1$ , the **direction of the  $0^\circ$  fibres** of the laminate. It must not be normal to the surface anywhere (error `SOR IS NORMAL TO SURFACE` ). Use one global direction for a part: the span for a wing, the axis for a fuselage, a tube or a cylinder.

The frame of a point is  $(e_1, e_2, e_3)$  with  $e_3$  the normal of the shell ( $e_2$  the tangent of the beam axis), so stresses and strains of the output ( `LOC` frame) are in the axes of the material.

## 13.3 Directors

A shell node has a **director**, the unit vector along which the thickness is measured (the normal). By default the director of a node is the average of the normals of the elements that share it, **limited to the elements within  $40^\circ$  of the normal of the element**: on a smooth surface the director is continuous, on a true edge (two walls of a box, a rib on a skin) every element keeps its own normal.

The normals follow the order of the nodes:  $a_1 \times a_2$ , with  $a_1$  the direction of the first side of the element (from node 1 towards node 2 for `Q4`, `Q9`, `Q16` ) and  $a_2$  the second one. **Number the nodes of the shells of a surface so that the normals point to the same side**; the program warns ( `SHELL NORMALS ARE OPPOSITE AT A SHARED NODE` ) when two elements with opposite normals share a node.

The polynomial surface is not exactly the real surface: on a cylinder discretised with quadratic elements of  $45^\circ$  the director of the polynomial surface is a few degrees off the radial direction (about  $0.3^\circ$  with  $15^\circ$  elements). If the exact normal is known (CAD, analytic surface) give it in the optional file `DIRECTORS.dat` :

```
8                                number of directors
1  0.0 0.0 1.0                 node_id Vx Vy Vz  (any length)
2  0.7071 0.0 0.7071
...
```

The directors of the file replace the average for the elements within  $40^\circ$  of them; the other nodes keep the automatic ones. The effect on the results is small (below  $10^{-3}$  on the pinched cylinder), the effect on the positions of the thickness points of a model with `LE` expansion is visible: a point load on an `LE` thickness node is found by its coordinates and needs the exact director.

### 13.4 Shear and membrane locking

`ANALYSIS.dat` selects the treatment per family (beam, plate, solid) as before. For curved elements:

| Token          | Curved beam                                                                  | Shell                                                                                |
|----------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| NONE           | compatible strains                                                           | compatible strains ( <b>locks</b> : do not use for thin shells)                      |
| MITC           | axial strain and the two shears tied at the $n - 1$ Gauss points of the axis | S9 : MITC9 (Bucalem-Bathe); S4 : MITC4 (transverse shears); S16 : compatible strains |
| REDI ,<br>SELI | <b>error</b>                                                                 | <b>error</b>                                                                         |

MITC (the default of the examples) is the one to use. The pinched cylinder below reaches 0.98 of the reference with MITC9 on  $8 \times 8$  elements per octant and 0.35 without tying on  $6 \times 6$ .

### 13.5 Joining the elements: shared nodes and node-dependent kinematics

Elements are connected by **shared nodes**, with **no multipliers and no constraint equations**: the displacements are the global components of the nodes, so two elements that share a node are continuous. The only rule is the existing one: *every*

*element that shares a node sees the same number of expansion terms at that node.* Within one element each node may have its own kinematics (node-dependent kinematics, KINEMATICS.dat ), so the order can change from node to node.

| Joint                                                           | How                                                                                                                                                                     |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| shell – shell, flat or curved, also at an edge (box, rib, spar) | share the nodes, same thickness expansion order ( <code>TE 2</code> on both sides works for walls at 90°: square tube within 1.3 % of beam theory)                      |
| shell – shell with another thickness                            | the zones have different expansion meshes; <code>TE</code> nodes of the same order can be shared (the faces step, the mid-surface is continuous)                        |
| beam – shell (stringer on a skin)                               | share the nodes and give them the <b>order 0</b> ( <code>TE 0</code> : only $u, v, w$ ) in the beam and in the shell; the other nodes of both keep their own kinematics |
| curved beam – straight beam                                     | share the nodes; the section meshes may differ with <code>TE</code> if the term counts agree                                                                            |

**Edges (kinks).** At a node shared by shell elements whose normals differ (more than the 40° feature angle, e.g. the corner of a box, a rib on a skin) the program does not share the first-order Taylor term as a Cartesian vector (that would tie the thickness derivative of walls with different thickness axes and lock the bending: a box came out 5 to 10 times too stiff) but expresses it as the rotation  $\omega$  of the node acting on the director of each element,  $q_1 = \omega \times V$ . The joint is then a rigid edge: displacements and rotations are continuous. Elements of opposite normals at the same node (a knife edge, the trailing edge of a wing) share the term with the sign of the thickness axis. This needs no input, but it is available for `TE` kinematics of order 1 or 2 (not for `LE`) and **every shell at the node must be a general element** (name them S4/S9/S16).

A node with `TE 0` has no rotation: the skin is rotationally restrained at the node, which is what a stiff stringer does and the stringer is a rod (axial and shear stiffness). **Use `TE 0` only at isolated nodes:** a *line* of `TE 0` nodes across the bending direction of a skin (a rod along a whole rib, a ring) forces the slope to zero and the structure is several times too stiff. The test `skin + stringer` gives the exact axial stiffness  $E(tb + A) / L$  to  $2 \cdot 10^{-5}$ .

## 13.6 Boundary conditions and loads

- **D-PLANE** on a **shell** selects the nodes by their **mid-surface point**: a plane through a shell node constrains the whole thickness of the node (all the terms), even if the director is a fraction of a degree off the plane. For a **beam** the section points are used (a plane can cut the section, as before).
- **F-POINT** is applied at the node and at the expansion node whose position (node + offset along the triad of the node) coincides with the point; for a **TE** node the load goes to the terms through the basis at the point.
- The surface loads **Q-PLANE**, **Q-SUN**, **Q-CONV** are **not available** with curved elements (error).
- Analyses: 101, 103, 104 and 106 (also with the temperature and potential fields); **105 and 108 stop with an error** when the model has curved elements.

## 13.7 Output

The points of `POSTPROCESSING.dat` are located inside the curved elements (Newton iteration on the map) and the displacements, strains, stresses (and, in the multiphysics models, potential and temperature) are written as for the other elements; the `LOC` frame is the frame of the point. The VTK output ( `PARA` ) draws the nodes of the expansion along the directors.

## 13.8 Examples (all in `TESTS/CURVED/curved_tests.py`)

| Problem                                 | Reference                               | Result                                                                 |
|-----------------------------------------|-----------------------------------------|------------------------------------------------------------------------|
| quarter ring cantilever, tip force, CB4 | energy solution (bending, axial, shear) | 0.9999 with 4 elements ( $\nu = 0$ )                                   |
| the same with $\nu = 0.3$               | same                                    | 0.995 with 8 elements (the 3D clamp is slower, as for a straight beam) |
| ring, tip couple                        | $Mx / I$                                | stress within 0.1 %                                                    |
| free ring (103)                         | six rigid-body modes                    | first elastic frequency 4.9037 Hz, converged to $10^{-5}$              |
| flat plate <code>s9</code>              | <code>Q9</code>                         | identical to the last digit (TE, LE, MITC, rotated ply)                |

| Problem                                                             | Reference                                | Result                       |
|---------------------------------------------------------------------|------------------------------------------|------------------------------|
| pinched cylinder, S9, TE 2, LE                                      | MacNeal-Harder<br>$1.8248 \cdot 10^{-5}$ | 0.98 (8 × 8), 1.00 (12 × 12) |
| cantilever circular tube                                            | beam theory                              | 0.990                        |
| cantilever square tube (shared corner nodes, rotation at the edges) | beam theory                              | 0.995 to 1.03                |
| skin + stringer joined at TE 0 nodes                                | $E(tb + A) / L$                          | $2 \cdot 10^{-5}$            |

## 13.9 Limits

- No curved beams or shells in 105 (buckling) and 108 (nonlinear); no surface loads on them.
- No arbitrary surface or body loads: loads are nodal ( F-POINT ), as before.
- A beam shares nodes with a shell only at the nodes where both have the same number of terms (order 0 in practice).
- S16 has no tying table (compatible strains).
- The directors of a very coarse surface are inaccurate: use DIRECTORS.dat or refine.

# Chapter 14 - Complete-aircraft demonstrator (analysis 103)

EXAMPLES/AIRCRAFT/make\_aircraft.py writes the input of a twin-engine transport aircraft of the 737 class (38 m, span 36.6 m) and MUL2\_V3.exe computes its free-vibration modes with analysis 103. It is a *dimensional* demonstrator: the sizes, the thicknesses and the masses are of the right order, not those of a real aircraft. Its purpose is to show that 1D, 2D and 3D elements, curved and flat, are put together with nothing but shared nodes and node-dependent kinematics.

```
python EXAMPLES/AIRCRAFT/make_aircraft.py --mesh fine --out RUN --modes 40
cd RUN
MUL2_V3.exe INPUT
python EXAMPLES/AIRCRAFT/plot_modes.py . 4 5 6 8
```

## 14.1 The model

| Part                                      | Elements                                    | Notes                                                                                                                              |
|-------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| fuselage skin (nose, cylinder, tail cone) | S9 shells, curved                           | surface of revolution; the thickness comes from the expansion mesh of the zone ( EXP_MESH_nn ), as for plates                      |
| frames, floor, floor beams                | S9 (web and flange), S9 plate and S9 strips | the floor beams carry the payload mass in their density                                                                            |
| wings, horizontal and vertical tails      | S9 skins, spars and ribs                    | the skins of the lifting surface share the nodes of the fuselage at the root; skin, spar and rib share the nodes along their edges |
| leading- and trailing-edge fairings       | S9                                          |                                                                                                                                    |
| engine pylon                              | four S9 plates (box)                        |                                                                                                                                    |
| engine                                    | H8 sheared solid blocks                     | joined to the pylon at isolated TE 0 nodes                                                                                         |
| tie rods (pylon / wing skin)              | CB3 straight and curved                     | TE 0 at the two end nodes                                                                                                          |

The fine mesh has 14 433 nodes, 3 706 shells, 4 curved beams, 48 solids and 129 093 degrees of freedom. The materials are aluminium (skin, frames, ribs), an orthotropic carbon-epoxy ply (ORT-M, laminations 5/6, available for the skins), an equivalent material for the engines and two aluminium variants with a larger density that carry the **mass of the fuel (wing webs) and of the payload (floor beams)**. No mass element is used: the masses are smeared in the densities.

All the families are joined with **shared coincident nodes** (no multipliers, no constraint equations). The shells use `TE 2` at every node, also at the edges between skin, spar and rib (the first-order term of the node is the rotation of the node there, Chapter 13). The beams and the solids are joined at nodes with `TE 0`.

## 14.2 Modelling rules that the demonstrator confirms

1. **Name the shells s9** (or `s4`, `s16`): the elements are then general elements and the edges between shells work as rigid edges. Shells of the same family that meet at a node must all be general.
2. **A `TE 0` node is a point, not a line.** A row of `TE 0` nodes along a rib or a ring suppresses the bending slope of the skin and makes the structure several times too stiff. The first versions of the demonstrator had stringers and longerons modelled as rows of `CB3` rods, and the clamped wing came out 4 to 6 times stiffer than beam theory; the stiffeners are now represented by the thicker skin (the bending of the box is carried by the skins), and the rods are used only for the tie rods of the engines.
3. A concentrated mass can be given as a larger density of the part that carries it (floor beams, fuel webs).

Check of the wing: the clamped wing box (uniform section, no fairings and no engines) with a tip load of 10 kN deflects 13.9 mm, against 13.8 mm of the beam theory ( $PL^3 / 3EI$  with the thin-walled section).

## 14.3 Results (fine mesh, 40 modes, 122 s wall, ARPACK 60 s)

The model is free: the first six modes are the rigid-body modes (three of them come out as numerically negative eigenvalues of the order of  $10^{-6}$  and are discarded with a warning). The first elastic modes are:

| Mode | f [Hz] | Character                                            |
|------|--------|------------------------------------------------------|
| 4    | 0.89   | vertical bending of wings and fuselage, with the fin |
| 5    | 1.55   | lateral bending of the fuselage and of the fin       |
| 6    | 2.25   | wing bending, symmetric                              |
| 7    | 2.61   | vertical, wing and tail                              |
| 8    | 2.63   | wing bending, antisymmetric, with fin                |
| 9    | 3.82   | lateral, fin                                         |
| 10   | 3.95   | wing in-plane                                        |
| 11   | 4.76   | outer wing                                           |
| 14   | 7.45   | fuselage                                             |

From about 9 Hz the spectrum is a mixture of global modes and of local modes of the floor and of the fuselage skin between frames (the table printed by `plot_modes.py` gives, for every mode, the fraction of the points that move and the share of the modal displacement in fuselage, wings and fin). The pictures `aircraft_mode_NN.png` show the plan and side view, the colour is the modulus of the modal displacement and the points are moved by the mode.

## 14.4 Static analysis (101)

`--static F` writes the same model as an analysis 101: fuselage clamped at the nose ring (D-PLANE at  $x = 5$  m) and a vertical force  $F$  at the tip of each wing. With  $F = 50$  kN the maximum displacement is 0.34 m, the two tips deflect 0.3301 m (symmetric to  $5 \cdot 10^{-5}$ ) and the run takes 27 s (assembly 4.8 s, solution 2.5 s). The results are in `EXAMPLES/AIRCRAFT/RESULTS_STATIC`.

## 14.5 Time

Assembly of  $K$  and  $M$ : 4.9 s; ARPACK, 40 modes: 60 s; the writing of the VTK/GMSH output of 40 modes is 56 s. The medium mesh (7 600 nodes, 68 000 DOF) takes 34 s for 30 modes, the coarse one a few seconds.

## 14.6 Limits

- Idealised aircraft: the sizes are plausible, the stiffnesses of the stiffeners and the masses are not those of a real design (the first bending frequency of a real aircraft with fuel is higher).
- No stringers modelled as separate beams over the skin: a stringer is a row of beam nodes with  $\theta = 0$  and the skin would be locked along it (Chapter 13, limits). A full stringer model needs a beam element with the rotation of the skin, which is a future extension.
- Curved elements are available in 101, 103, 104 and 106; the static and the dynamic response of the same model can therefore be computed, buckling and the nonlinear analysis cannot.

# Quick reference

## A.1 Files and records

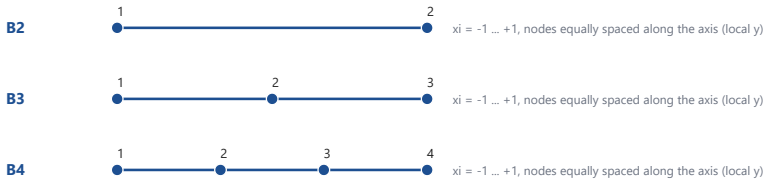
**Table A.1.** One-page summary of the input files.

| File               | Record          | Syntax                                                                                                                                                            |
|--------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANALYSIS.dat       | analysis        | 101 or 103 ; line 2 number of modes; lines 3–5 NONE / REDI / SELI / MITC for beam, plate, solid; optional last line FIELDS MECH [THERMO] [PIEZ0] (physics solved) |
| NODES.dat          | node            | id X Y Z TE n or id X Y Z LE                                                                                                                                      |
| CONNECTIVITY.dat   | element         | TYPE id n1...nN versor section                                                                                                                                    |
| VERSORS.dat        | versor          | VECTOR id vx vy vz                                                                                                                                                |
| MATERIAL.dat       | header          | n_materials n_records                                                                                                                                             |
|                    | isotropic       | ISO-M id E nu rho                                                                                                                                                 |
|                    | orthotropic     | ORT-M id EL ET EZ nuLT nuLZ nuTZ GLT GLZ GTZ rho                                                                                                                  |
|                    | anisotropic     | MAT-C id c11...c66 rho                                                                                                                                            |
| LAMINATION.dat     | lamination      | LAM2 id material angle_y angle_z (degrees)                                                                                                                        |
| EXP_MESH_nn.dat    | section<br>node | id x y z (local coordinates)                                                                                                                                      |
| EXP_CONN_nn.dat    | sub-<br>element | TYPE id lamination n1...nN                                                                                                                                        |
| BC.dat             | constraint      | D-PLANE id A B C D ux uy uz (plane $AX + BY + CZ + D = 0$ ; N = free)                                                                                             |
|                    | load            | F-POINT id x y z Fx Fy Fz                                                                                                                                         |
| POSTPROCESSING.dat | field<br>output | PARA or GMSH, cell_nodes, GLB or LOC, s1...s9 (cell nodes 8, 20, 27)                                                                                              |
|                    | point           | PNT id x y z                                                                                                                                                      |
|                    | dumps           | KMAT MMAT FRCE UNKN ENRG                                                                                                                                          |

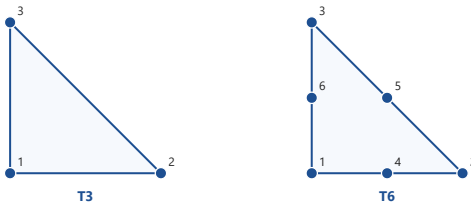
## A.2 Element node order

**Table A.2.** Node order of the elements.

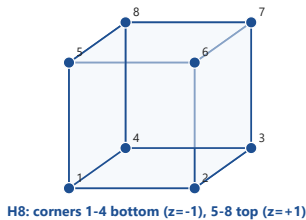
| Element    | Order                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------|
| B2, B3, B4 | along the axis, 1 ... N, equally spaced                                                                                        |
| Q4         | counter-clockwise from the corner ( - 1, - 1): 1 2 3 4                                                                         |
| Q9         | corners and mid-sides alternate counter-clockwise from ( - 1, - 1): 1 (corner) 2 (mid) 3 (corner) 4 5 6 7 8, <b>9 = centre</b> |
| Q16        | 12 boundary nodes counter-clockwise from the corner ( - 1, - 1) (4 per side, corners 1, 4, 7, 10), then interior 13–16         |
| T3 / T6    | corners 1 2 3; T6: 4 = mid 1–2, 5 = mid 2–3, 6 = mid 3–1                                                                       |
| H8         | bottom face 1–4 counter-clockwise, top face 5–8 above them                                                                     |
| H27        | corners 1–8 as H8, then edges 9–20, faces 21–26, centre 27                                                                     |



**Figure A.1.** Beam elements.



**Figure A.2.** Triangles.



#### H27 node classes

1-8 corners (same order as H8)

9-20 mid-edge nodes

21-26 mid-face nodes

27 centre node

Natural coordinates ( $\xi, \eta, \nu$ ) in  $[-1, 1]^3$ .

H20 has corners and mid-edge nodes only.

P6: two triangles (1-3 and 4-6); T4/T10: tetrahedra.

**Figure A.3.** Hexahedra.

## A.3 Frames

**Table A.3.** Frames at a glance.

| Element | Local $y$                                | Local $z$                                                       | Local $x$                  |
|---------|------------------------------------------|-----------------------------------------------------------------|----------------------------|
| beam    | the axis (first $\rightarrow$ last node) | $v$ made orthogonal to the axis                                 | $y \times z$               |
| plate   | $z \times x$                             | normal (node 1 $\rightarrow$ 2 $\times$ node 1 $\rightarrow$ 3) | $v$ projected on the plate |
| solid   | $Y$                                      | $Z$                                                             | $X$                        |

Material axes: with zero angles  $(T, L, Z) = (x, y, z)$ ; fibre direction is  $L$ .

## A.4 Typical numbers

**Table A.4.** Number of unknowns per structural node.

| Expansion                         | Terms        | DOF per node  |
|-----------------------------------|--------------|---------------|
| beam TE 1, TE 2, TE 3, TE 4       | 3, 6, 10, 15 | 9, 18, 30, 45 |
| beam LE with $m$ section nodes    | $m$          | $3m$          |
| plate TE $n$                      | $n + 1$      | $3(n + 1)$    |
| plate LE with $m$ thickness nodes | $m$          | $3m$          |
| solid                             | 1            | 3             |

## A.5 Command line

```
MUL2_V3.exe [input_directory]      # default: PATH_input.dat, then INPUT
set OMP_NUM_THREADS=8              # threads
set MUL2_DEBUG=1                   # debug messages
```

## A.6 Order of the strain and stress components

$(xx, yy, zz, xz, yz, xy)$ , engineering shear strains. Displacements in `POST_POINT.dat` are global.